

Machine Learning Driven Advances in Molecular Dynamics of Bulk and Interfacial Aqueous Systems

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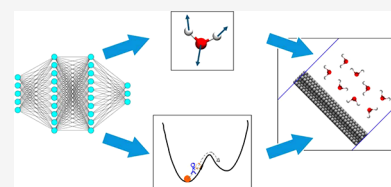
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ABSTRACT: Molecular dynamics (MD) simulations have been widely applied to investigate various physical and chemical processes in aqueous and interfacial environments, which are crucial for the design of energy materials and for understanding several chemical processes at the heart of life itself. However, the applicability of MD simulations has been constrained by several inherent challenges, including the accuracy of force fields, limitations in simulation size and time scales. One promising solution to these challenges is the integration of machine learning (ML) methods, both for improved description of the nature of interactions in aqueous systems as well as for enhanced sampling. In this review, we discuss the principles, implementation, and applications of ML force fields (MLFFs) and ML enhanced sampling methods to the study of aqueous, interfacial systems. We discuss five key categories of applications that use MLFFs, ML-enhanced sampling, and ML-driven data analytics. We first discuss how MLFFs are enabling quantum level accuracy at classical level cost for large scale simulations of complex aqueous and interfacial systems, and then we highlight how coupling them with enhanced sampling and advanced data analytics, especially graph based approaches for featurizing such systems, can be used both for enhancing simulations and for understanding them by yielding reliable low dimensional reaction coordinates that improve the interpretation of high dimensional MD data. The discussed applications include investigations into the structure and dynamics of bulk water and aqueous interfaces, proton transfer, catalysis, phase transitions, and the prediction of vibrational spectra. In each case, we highlight how ML-based methods enable simulations that were previously computationally prohibitive and provide new physical insights into aqueous solutions and interfaces. For instance, MLFFs allow nanosecond-scale simulations with thousands of atoms while maintaining quantum chemistry accuracy. Additionally, ML-enhanced sampling facilitates the crossing of large reaction barriers and enables the exploration of extensive configuration spaces. Moreover, ML models trained on simulation data uncover previously overlooked factors, such as the role of solvent dynamics in phase transitions. The combination of MLFFs with enhanced sampling techniques makes the calculation of high-dimensional free energy surfaces feasible, significantly improving our understanding of chemical reactions. Finally, we discuss the current challenges in this field and outline potential future research directions to further advance the integration of ML in MD simulations.



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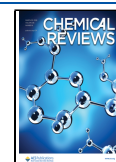
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1. INTRODUCTION

Water, one of the most ubiquitous substance on earth, is essential to human life and plays a critical role in countless chemical and physical processes, including phase transitions, catalysis, material design, protein folding, adsorption of pollutants from the air, clean energy production, origins of life, drug design, etc.^{1–9} Besides bulk water, often it is water in the vicinity of an interface that plays a critical role in science and engineering problems.^{10–15} To give a few examples, electrochemical catalysis related processes such as hydrogen evolution reactions depend on water dissociation and the adsorption of hydrogen atoms onto metal electrode surfaces.¹⁶ The adsorption and hydrolysis of N₂O₅, a key contributor to acid rain, occur at water/air interfaces.¹⁷ In biochemistry, water governs the function of proteins by forming hydrogen bonds that stabilize their folded structure, especially near hydrophobic or partially solvated surfaces where the interplay between water structuring and surface chemistry critically modulates folding and function.¹⁸

Numerous experimental techniques have been developed and applied to investigate structure and dynamics of bulk and interfacial water, including but not limited to infrared, Raman, sum frequency generation (SFG), atomic force microscopy, X-ray reflectivity, scanning tunneling microscopy, and X-ray adsorption spectroscopy.^{19–27} However, many of these experimental techniques provide only indirect spectra, and proper hypothesis and modeling are required to interpret experimental observation in order to recover the true structure and dynamics at the microscopic scale of interest.¹⁰ Additionally, impurities present under experimental conditions may introduce artifacts and lead to misinterpretation of the experimental observations.^{28,29} Molecular dynamics (MD) simulations, on the other hand, can reveal microscopic structural information at

the atomic scale and dynamics on the picosecond time scale. In recent years, MD simulations have gained increasing attention due to advances in computational power. Besides supporting and interpreting experimental observations, MD simulations can also be applied to investigate chemical processes that are difficult to study experimentally.

Despite the utility of MD simulations, their inherent limitations hinder potential broader applications. The first major challenge in MD simulations is accurately reproducing energy and forces of atoms from given configurations. While potential energy surfaces can be computed using high-level quantum chemistry methods, the associated high cost is usually not affordable for MD simulations, which commonly employ two approaches: *ab initio* MD and classical MD. *Ab initio* MD simulations are often carried out at the density functional theory (DFT) level to reduce the high computational cost of other high-level quantum methods. But they still remain prohibitively expensive for large-scale simulations. In contrast, in classical MD simulations, forces are calculated using simplified analytical functions based on atomic geometries, such as distances, angles, and dihedral angles. The approach, known as force fields (FFs), is designed for fast computation. However, classical MD simulations often neglect bond breaking and formation, as well as more complex atomic interactions involving electronic interactions, including not limiting to the dipole and polarization of atoms. Plus, vibrations are predefined or missing in classical MD, making it difficult to estimate the vibrational spectra.³⁰ These limitations also make classical MD unsuitable for modeling chemical reactions. The second major challenge is the limited time scale accessible in MD simulations. Typical *ab initio* MD simulations span a few picoseconds and contain hundreds of atoms, whereas modern classical MD simulations can reach time scales of several microseconds or even milliseconds for systems approaching millions of atoms. Many of the macroscopic physical and chemical processes relevant to modeling bulk and interfacial aqueous systems are much slower, and occur on longer time scales. Unbiased MD simulations follow the Boltzmann distribution but often get trapped in metastable states, limiting exploration of high-barrier events and wasting computational resources. Moreover, the resulting high-dimensional data require dimensionality reduction for effective analysis.³¹

Many physics-based methods have been developed to address the challenges discussed above. Advanced force fields that include bond structure have been developed, such as the Reactive Force Field (ReaxFF),³² MB-pol,³³ and the Multiscale Reactive Molecular Dynamics Simulations.³⁴ These methods enable the modeling of certain chemical reactions in aqueous phases, such as proton transfer. Polarizable FFs have also been developed to improve the accuracy of water models.^{35,36} However, the development of these models has predominantly focused on pure water and simple ions. When there are many species in simulations, the transferability and compatibility of FFs for multiple species limit their applicability to more complex systems, and much work remains to be done. To overcome the challenge of sampling rare events, enhanced sampling methods have been introduced to extend the range of configurations explored in MD simulations. These approaches effectively accelerate simulations or increase the likelihood of observing low-probability configurations.³⁷ While the simulation trajectories obtained using enhanced sampling are not always physical, proper reweighting techniques can be applied to recover accurate reaction kinetics and free energy surfaces.^{38,39} Certain

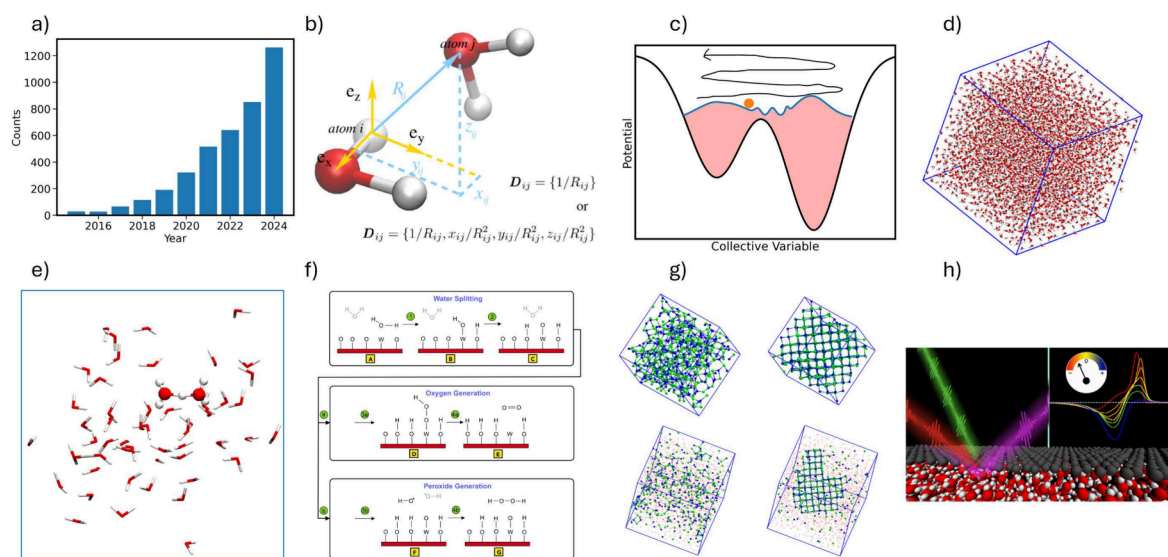


Figure 1. Different ways in which machine learning and molecular dynamics simulations have become popular and are integrated. (a) The number of papers found by searching the keywords “machine learning” and “molecular dynamics simulation” in the Web of Knowledge database for each year (as of February 2025). (b) Forces can be calculated using machine learning force fields. (c) Machine learning has been applied to enhance sampling in molecular dynamics simulations. (d)–(h) Applications of molecular dynamics simulations combined with machine learning include predicting and modeling: (d) behaviors of bulk water, (e) proton transfer, (f) catalysis, (g) phase transitions, and (h) vibrational spectra. Figure (b), (f), (g), and (h) are reprinted from ref.⁵¹ (Copyright 2018, American Physical Society), ref.⁵² (Copyright 2024, Elsevier), ref.⁵³ (Copyright 2024, American Chemical Society), and ref.⁵⁴ (Copyright 2025, American Chemical Society), respectively.

enhanced sampling methods, such as umbrella sampling and metadynamics, require predefined collective variables (CVs) to guide the sampling process. However, human-intuitive CVs are not always sufficient to approximate the true Reaction Coordinates (RCs), which can limit the effectiveness of enhanced sampling methods.

To address the remaining challenges discussed above, machine learning (ML) has garnered growing attention, driven by advances in computer hardware and software that have significantly accelerated ML calculations. Especially when following the principles of statistical mechanics, ML methods can be employed to identify RCs or slow modes, offering deeper insights into system dynamics and improve the reliability of enhanced sampling. ML is being used not just to enhance sampling but also to achieve DFT-level accuracy at computational costs comparable to classical MD. For this, ML-based force fields (MLFFs) have been developed and applied to a variety of systems.

Integrating ML into MD simulations is particularly important for studying aqueous and interfacial systems due to the inherent complexity of these environments. Explicit modeling of solvent water typically requires a large number of water molecules, especially at relatively low solute concentrations. Moreover, the long-range electrostatic interactions between charged species and water demand simulation box sizes on the order of several nanometers to achieve convergence. Modeling many interfacial processes requires the inclusion of electronic effects. These include, to name a few: the polarization of solid surfaces, the breaking and formation of chemical bonds during proton transfer, and vibration of water. MLFFs are introduced to achieve DFT-level accuracy at a reasonable computational cost for aqueous interfacial systems. To investigate interfacial chemical catalysis for clean energy applications, it is essential to calculate high-dimensional free energy landscapes of reaction pathways, which can only be achieved through carefully designed enhanced sampling MD simulations.

Our intention in writing this review is to survey broadly the wide domain of computational all-atom and electronic resolution studies that are being carried out in aqueous solutions and interfaces. These topics have gained significant attention in recent years, and numerous review papers have been published discussing either MLFFs or ML-based statistical mechanics (Figure 1a),^{40–50} but rarely together in the same review. In this expansive and hopefully comprehensive review, we specifically focus on MD simulations of aqueous and interfacial systems, with an emphasis on their applications in energy and materials science. We will demonstrate how the combination of machine learning FFs, enhanced sampling and data analysis is advancing the investigation into this field. We will first discuss the theoretical foundations of MLFFs (Sec. 2, Figure 1b) and ML enhanced sampling (Sec. 3, Figure 1c). This will be followed by examples illustrating the integration of ML and MD in aqueous and interfacial environments, including the structure and dynamics of bulk and interfacial water (Sec. 4.1, Figure 1d), proton transfer (Sec. 4.2, Figure 1e), catalysis (Sec. 4.3, Figure 1f), phase transition (Sec. 4.4, Figure 1g), and prediction of vibrational spectra (Sec. 4.5, Figure 1h). Finally, we will discuss our perspective on the current state and future opportunities (Sec. 5) in this rapidly evolving area of research.

2. ML POTENTIALS AND FORCE FIELDS

2.1. Overview

MLFFs have gained increasing popularity in recent years due to the growing demand for simulations that require DFT-level accuracy and capacity for modeling chemical reactions but at classical-level time scales and size.^{55–60} As discussed in the Introduction section, simulations of aqueous and interfacial systems typically involve a relatively large number of molecules and, in certain applications, require the inclusion of electronic effects. MLFFs are well-suited to address both demands, offering quantum-level accuracy and information at a reasonable

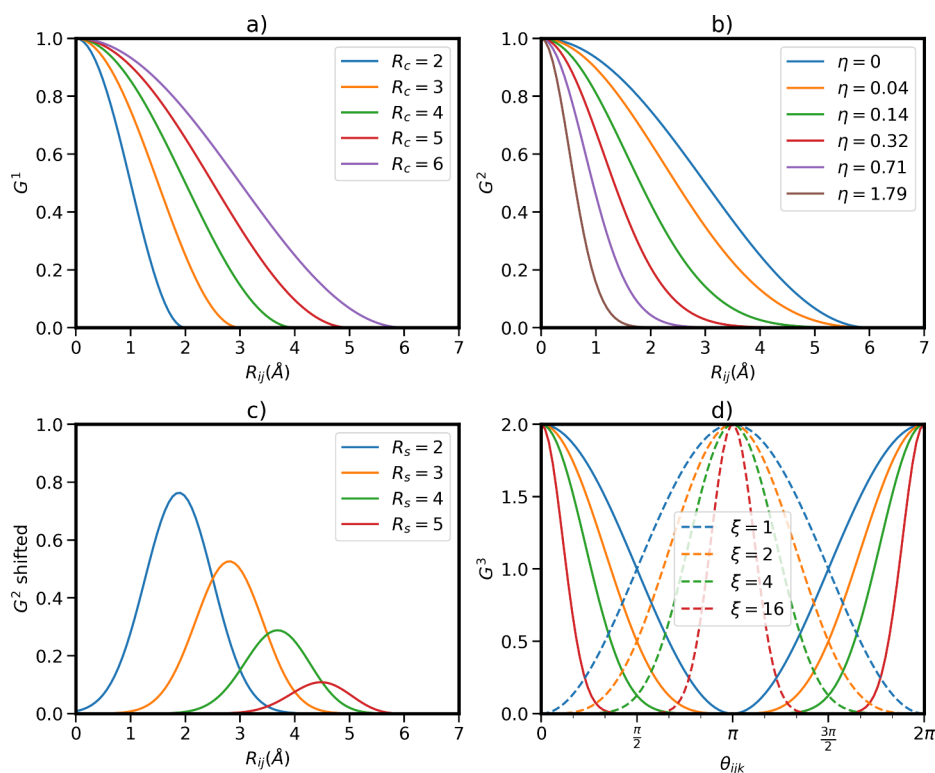


Figure 2. Simplified version of Behler-Parrinello symmetry functions. G^1 and G^2 represent racial symmetry functions, while G^3 is an angular symmetry function. (a) G^1 is eq 1, with $R_c = 6$ Å. (b) G^2 is eq 4, with $R_s = 0$ and $R_c = 6$ Å. The unit of η is Å². (c) Shifted G^2 with $\eta = 1.2$ Å² and $R_c = 6$ Å. (d) G^3 is eq 5. Each color represents a ξ . Solid and dashed lines represent $\lambda = 1$ or -1 , respectively. For simplicity, radial terms are neglected in the panel, i.e., $R_{jk} = R_{ik} = 0$. Figures are plotted based on ref.⁶¹ (Copyright 2015, John Wiley and Sons).

computational cost. The primary goal of MLFFs is to learn the potential energy surface (PES) of a molecular system, from which the forces between atoms can be derived. Beyond forces, MLFFs can also predict properties such as dipole moments, charge distributions, and electronic states of molecules. In this section, we will discuss MLFFs based on descriptors (Sec. 2.2), end-to-end MLFFs (Sec. 2.3), and physics-based MLFFs (Sec. 2.4). In addition, we have also included discussion on MLFFs with long-range interactions (Sec. 2.5), coarse-grained MLFFs designed to model macroscopic processes (Sec. 2.6), and MLFF foundation models (Sec. 2.7).

2.2. Descriptor-Based MLFFs

2.2.1. Descriptors. The intrinsic properties of the PES exhibit invariance to translation, rotation, reflection, and permutation of identical atoms. ML models must inherently account for this invariance, rather than relying on input data to explicitly encode them. In practice, the PES in simulations is calculated from all atomic coordinates, however, atom-wise Cartesian coordinates are not ideal inputs for MLFFs because they break permutation invariance, i.e. swapping two atoms alters the input. To address this challenge, modern MLFFs utilize descriptors derived from simulation configurations that inherently preserve these symmetries. Additionally, since the coordination number of each atom in simulations can vary and the number of neighboring atoms required to describe the local environment differs, descriptors must be sufficiently flexible to adapt to input data with varying atomic environments. Here we describe some such popular descriptors.

Atom Centered Symmetry Functions (ACSFs). Symmetry functions are specifically designed to transform atomic coordinates into descriptors that describe local atomic environ-

ments. In MLFFs, symmetry functions eliminate the dependence on atom order, ensuring permutation invariance. Consequently, atoms with identical environments within a certain cutoff will have the same values or vectors for their symmetry functions.

Originally, Behler and Parrinello proposed the following types of symmetry functions:^{61–63}

$$f_{c,1}(R_{ij}) = \begin{cases} 0.5 \left[\cos\left(\frac{\pi R_{ij}}{R_c}\right) + 1 \right], & \text{if } R_{ij} \leq R_c \\ 0, & \text{if } R_{ij} > R_c \end{cases} \quad (1)$$

or

$$f_{c,2}(R_{ij}) = \begin{cases} \tanh^3 \left[1 - \frac{R_{ij}}{R_c} \right], & \text{if } R_{ij} \leq R_c \\ 0, & \text{if } R_{ij} > R_c \end{cases} \quad (2)$$

Both symmetry functions take pairwise distance between atoms i and j (R_{ij}) as input, where the function value decreases continuously from 1 to 0 as the distance approaches a predefined cutoff R_c (Figure 2a). The continuous nature of the symmetry function ensures differentiability with respect to R_{ij} , enabling the calculation of forces by taking derivatives of the PES with respect to atomic coordinates. The radial symmetry descriptor of an atom is defined as the summation over all neighboring atoms within the cutoff distance:

$$G_i^1 = \sum_{j=1}^{N_{\text{atom}}} f_c(R_{ij}) \quad (3)$$

Here, G_i^1 produces a single scalar value. For a more comprehensive characterization of the atomic environment, multiple cutoff values can be selected and computed to construct a descriptor vector that describes different pairs of atoms. An improved version of the symmetry function introduces a Gaussian function to smoothen the cutoff region (Figure 2b) and a shift on the center of symmetry functions (Figure 2c):

$$G_i^2 = \sum_{j=1}^{N_{\text{atom}}} e^{-\eta(R_{ij}-R_s)^2} f_c(R_{ij}) \quad (4)$$

where η represents the width of Gaussian functions, tuning how rapidly the symmetry function G^2 decays with distance. R_s specifies the shift of the Gaussian center, allowing the symmetry function to focus on specific regions around the center atom. One can also define the angular symmetry function that captures three body interaction:

$$G_i^3 = 2^{1-\xi} \sum_{j \neq i} \sum_{k \neq i, j} [(1 + \lambda \cos \theta_{jik})^\xi e^{-\eta(R_{ij}^2 + R_{jk}^2 + R_{ik}^2)} f_c(R_{ij}) f_c(R_{jk}) f_c(R_{ik})] \quad (5)$$

where θ_{jik} is the angle among atoms i , j , and k , i represents the center atom, ξ is a normalization parameter. Similar to radial symmetry functions, the width and center angles of these angular symmetry functions can be adjusted to focus on specific angular regions (Figure 2d).

Smooth Overlap of Atomic Positions (SOAPs). The Smooth Overlap of Atomic Positions (SOAPs) descriptors can also be used to describe the local similarity of atomic environments and have been applied in enhanced sampling MD to study phase transitions.⁶⁴ For each atom i , the density of its local environment is expressed as a Gaussian kernel on its center:

$$\rho^{i,s}(\mathbf{r}) = \sum_j \delta_{ss_j} \exp\left[-\frac{|\mathbf{r} - \mathbf{r}_j|^2}{2\sigma^2}\right] f_c(R_{ij}) \quad (6)$$

where s represents a specific species, δ_{ss_j} is the Kronecker delta for species s and s_j . To preserve rotational invariance, eq 6 is expanded using spherical harmonics function Y_{lm} and orthogonal radial basis functions:

$$\rho^{i,s}(\mathbf{r}) \approx \sum c_{nlm}^{i,s} G_n(r) Y_{lm}(\hat{\mathbf{r}}) \quad (7)$$

where n , l , and m are the quantum numbers in quantum chemistry to describe the degree and order in spherical harmonics functions. $\hat{\mathbf{r}}$ is the projection of $\mathbf{r}$ onto the unit sphere. c denotes the coefficients:

$$c_{nlm}^{i,s} = \int d\mathbf{r} G_n(r)^* Y_{lm}(\hat{\mathbf{r}})^* \rho^{i,s}(\mathbf{r}) \quad (8)$$

The SOAP descriptor is defined as the power spectrum of these coefficients:

$$p_{nn'}^{i,ss'} = \frac{1}{\sqrt{2l+1}} \sum_m (c_{nlm}^{i,s})^* c_{n'l'm'}^{i,s'} \quad (9)$$

where i represents the central atom, s and s' represent different species or elements. n and n' represent different radial regions. Finally, the SOAP value, similarity of atom environments

between A and A' , is defined as the inner product of atoms' smooth density:

$$k(A, A') = (\hat{p}^i \hat{p}^{i'})^\zeta \quad (10)$$

where $\hat{p}$ is the normalized power spectrum, $\hat{p}^i = p^i / \sqrt{p^i p^i}$, ζ is often 2 or 4 to tune the sharpness of k . The value of $k(A, A')$ lies between 0 and 1, representing dissimilarity and equality, respectively.

In summary, the SOAP descriptor characterizes atomic environments using neighbor densities expanded in radial and angular basis functions, ensuring rotational and permutational invariance. The power spectrum of these expansions serves as the descriptor, and the normalized dot product of SOAP vectors provides a measure of similarity between the environment of two atoms.

Atomic Cluster Expansion Descriptors. In the atomic cluster expansion (ACE) descriptors, the energy of an atomic system is modeled as the summation of many-body interactions:

$$E = \sum_n \frac{1}{n!} \sum V^{(n)} \quad (11)$$

For each atom i , the interaction of the order α with its K neighbors are described by $\Phi_{\alpha i} = \phi_{v_1}(r_{ij_1}) \phi_{v_2}(r_{ij_2}) \dots \phi_{v_K}(r_{ij_K})$, where ϕ are orthogonal basis functions containing radial basis function and angular basis functions.⁶⁵

2.2.2. Constructing MLFFs from Descriptors. After obtaining the descriptors of the chemical environment of each atom, the next step is to apply ML models to learn the structural information and estimate the energy contribution due to that particular atom. The total energy of the simulation system will be the summation of energy contributions from all individual atoms. Additionally, other properties relevant to MD simulations, especially the forces, can be derived from the gradient (derivative) of the energy with respect to atomic positions:

$$F_\alpha = F_{\alpha,s} + F_{\alpha,elec} = -\frac{\partial E_s}{\partial \alpha} - \frac{\partial E_{elec}}{\partial \alpha} \quad (12)$$

where the force on atom α is the summation of the short-range force $F_{\alpha,s}$ and the long-range electrostatic force $F_{\alpha,elec}$:

$$F_{\alpha,s} = - \sum_j \sum_{\mu}^{N_{\text{atom}} N_{\text{sym},j}} \frac{\partial E_j}{\partial G_{j\mu}} \frac{\partial G_{j\mu}}{\partial \alpha} \quad (13)$$

$$F_{\alpha,elec} = \sum_i \sum_{j \neq i}^{N_{\text{atom}} N_{\text{atom}}} \frac{Q_i}{R_{ij}} \left[\frac{1}{2} \frac{\partial R_{ij}}{\partial \alpha} \frac{Q_j}{R_{ij}} - \sum_k^{N_{\text{sym},i}} \frac{\partial Q_j}{\partial G_{ik}} \frac{\partial G_{ik}}{\partial \alpha} \right] \quad (14)$$

The short-range interaction entirely depends on the coordinates of the central atom and can be calculated directly from the symmetry function. In contrast, the electrostatic force is influenced by the partial charges assigned by neural networks. To account for the long-range charge–charge interactions, additional methods such as Ewald summation can be employed, using both the atomic coordinates and the assigned partial charges, similar to the approach used in classical MD simulations.

The final task is to train the neural network parameters so that the neural network potential can accurately reproduce the total energy of the system. In refs61–63, Behler and Parrinello pioneered the use of traditional feed-forward neural networks

(NNs) for this purpose. In this framework, the NN consists of an input layer, an output layer, and several hidden layers. The values of neurons in the next layer are calculated as

$$y_i^j = f_i^j \left(b_i^j + \sum_k^{N_{j-1}} a_{k,i}^{j-1} y_k^{j-1} \right) \quad (15)$$

where y_i^j is the value of node i at the j layer, f_i^j is the activation function, b_i^j is the bias term and a is the linear scaling coefficient. a and b are parameters need to be trained using the back-propagation algorithm. The training and validation processes often involve the following steps:

1. Perform quantum MD simulations at the DFT or other levels, and select a sufficient number of configurations that represent all possible states.
2. Iterate the following steps:
 - (a) Train MLFFs with the selected configurations, energies, and forces from previous step.
 - (b) Carry out trial MD simulations with the MLFF trained from the last step.
3. Stop the training when the error of the trained MLFFs is reasonable, and no unphysical configurations are observed in trial simulations.
4. Conduct the final production run of MD simulation using the trained MLFFs for further analysis.

In the Behler-Parrinello MLFFs, the symmetry functions are determined *a priori* through parameters such as the radial cutoff, Gaussian width, and shift, and these remain fixed throughout the training process and may lead to problems. This limitation has been addressed in the DeePMD MLFF, which allow for greater flexibility.⁵¹ In the DeePMD framework, atomic interactions are characterized using both radial and angular information defined within the local molecular frames. These descriptors are then fed into a deep neural network to compute the energy associated with each atom. During training, DeePMD effectively learns the hyperparameters that were originally required to be manually specified in Behler–Parrinello MLFFs. In addition, DeePMD employs deep neural networks with more layers and nodes, significantly enhancing its capacity to reproduce energy results with higher accuracy.

Another limitation of the Behler-Parrinello MLFF is that all atoms are treated using the same set of symmetry functions, making it challenging to perform simulations involving multiple types of atoms. To address this issue, Isayev and co-workers developed the Accurate Neural network engine for Molecular Energies (ANI) MLFFs.⁶⁶ The ANI model replaces symmetry functions with atomic environment vectors (AEVs) as descriptors. AEVs are specifically constructed for each type of element, significantly reducing the number of descriptors required while still accurately representing the molecular environment.

2.3. End-to-End MLFFs

End-to-end MLFFs take Cartesian coordinates as input and generate energy and forces directly from the data. Many of these models are based on graph neural networks (GNNs) or message-passing neural networks. In these models, atoms and their interactions are represented as nodes and edges, respectively. Nodes that are within a specified distance cutoff are treated as connected, allowing them to exchange node and edge features through their edges. Descriptors discussed in Section 2.2 can also be incorporated into graph models. In theory, any atom-wise features can be embedded for message

passing. The graph-based approach naturally captures the topological information on molecular configurations, and the type of atoms can be easily encoded in the node features.

In graph-based MLFFs, a node corresponding to atom i (σ_i) is generally modeled as

$$\sigma_i^{(t)} = (\mathbf{r}_i, z_i, \mathbf{h}_i^{(t)}) \quad (16)$$

where $\mathbf{r}$ represents the coordinates, z is the element, and $\mathbf{h}^{(t)}$ are node features at layer t . A message $\mathbf{m}_i^{(t)}$ is generated by performing a permutation-invariant pooling operation over the features of neighboring nodes within a cutoff distance. The message is then used to update the node features at the next layer:

$$\mathbf{h}_i^{(t+1)} = U_t(\sigma_i^{(t)}, \mathbf{m}_i^{(t)}) \quad (17)$$

where U_t is the update function that combines the current node state $\sigma_i^{(t)}$. Finally, the total energy of the system is obtained by mapping the updated node features to energy contributions, and are summed over all atoms.

One graph-based MLFF that has been successfully applied in MD simulations of aqueous solutions is the MACE model.⁶⁷ In the MACE model, messages are constructed using ν -body interactions, where ν is a tunable hyperparameter, as opposed to the two-body messages commonly used in traditional GNN-based MLFFs (eq 17). The edge features in MACE are built using permutation-invariant two-body interactions, represented as products of a series of radial basis functions and spherical harmonics. To construct higher-order features, tensor products of edge features are calculated and symmetrized. This approach allows MACE to naturally incorporate many-body interactions, overcoming the limitations of traditional GNNs that are restricted to two-body interactions. In the current implementation of MACE, a maximum of fourth order interactions have been included. As a result, the MACE MLFF demonstrates excellent scalability and the flexibility to support MD simulations involving multiple elements, making them particularly suitable for complex systems like aqueous solutions.

Another notable ML model is the Deep Tensor Neural Network (DTNN), which takes the distance matrix between atoms and their nuclear charges as inputs.⁶⁸ The distances are expanded using Gaussian regression models, and atom feature vectors are iteratively updated during training based on pairwise atom interactions. Although only pairwise interactions are explicitly included, the model is capable of learning higher-order interactions implicitly. Despite its advantages, early versions of DTNN had limitations, such as difficulties in modeling large molecules and the inability to predict forces. However, compared to descriptor-based methods, DTNN does not require predefined hyperparameters for symmetry functions, simplifying its framework. Later, Tkatchenko and co-workers introduced the SchNet MLFF, which overcomes some of DTNN's limitations, including force prediction.⁶⁹ SchNet utilizes continuous-filter convolutions instead of discretized, grid-based data, allowing it to model arbitrary atomic positions effectively. Its architecture consists of molecular representation layers, atom-wise layers, and interaction layers. Compared to DTNN, SchNet significantly reduces the mean absolute errors, offering improved accuracy and efficiency.

The long-range interactions can also be separately calculated in graph-based MLFFs. Unke and co-workers developed PhysNet, a machine learning force field capable of predicting energies, forces, dipole moments, and partial charges.⁷⁰ PhysNet

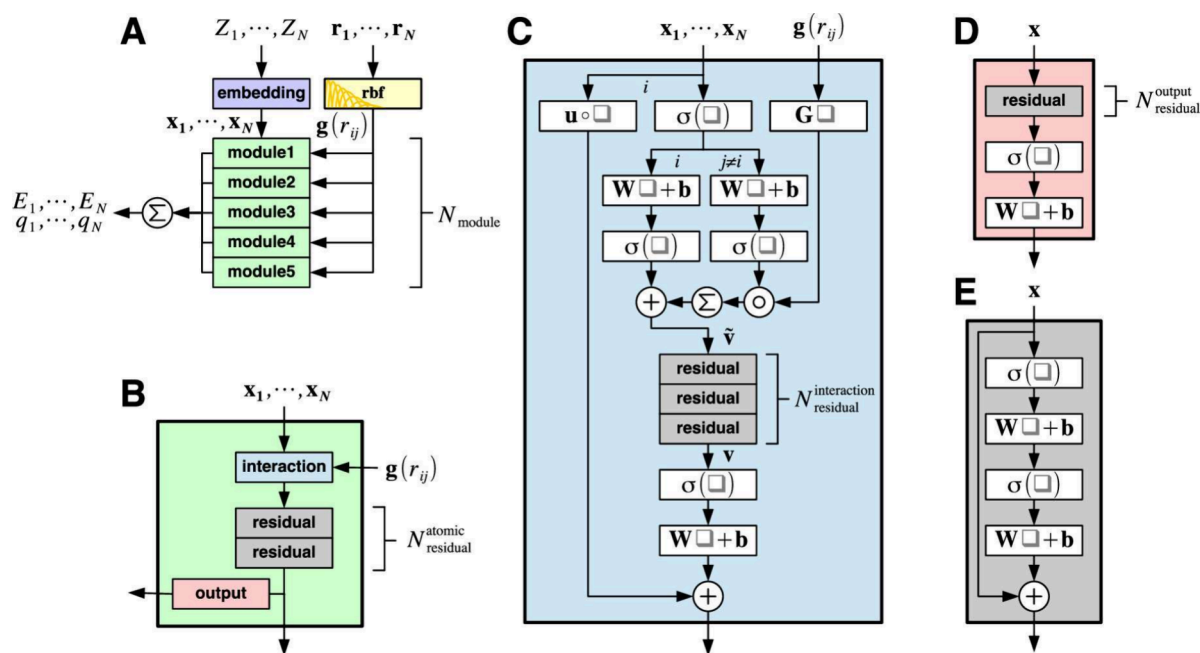


Figure 3. Architecture of PhysNet. (A) The structure of PhysNet. Z represents input charges and $\mathbf{r}$ represents atom coordinates. **rbf** represents the radial basis function to generate $\mathbf{g}$ functions. (B) The module block. The input contains atom features $\mathbf{x}$ and $\mathbf{g}$ functions. (C) The interaction block. (D) The output block. (E) The residual block, which contains a two-layer neural network. Figures are reprinted in ref.⁷⁰ (Copyright 2019, American Chemical Society).

consists of several key components, including an embedding layer, modular blocks, residual blocks, an interaction block, and an output block (Figure 3). These components are

1. The embedding layer (the purple box in Figure 3a) converts atomic elements and their positions into vector representations.
2. The module blocks (the green boxes in Figure 3b) process atomic features by combining the center atom's properties with those of all neighboring atoms within a defined distance cutoff.
3. Residual blocks enable the training of deeper neural networks by addressing vanishing gradient issues.
4. The interaction block (Figure 3c) learns the environment of a center atom using atomic features $\mathbf{x}$ and radial basis functions $\mathbf{g}(r_{ij})$. The neural network's weights ($\mathbf{W}$), biases ($\mathbf{b}$), and attention masks ($\mathbf{G}$) are trainable parameters.
5. The output layers (Figure 3d) use the learned atomic features to predict the energy contributions of individual atoms.

Notably, PhysNet can also predict partial charges on atoms, and the final total energy of the system is computed as

$$E = \sum_i E_i + k_e \sum_i \sum_{j>i} \hat{q}_i \hat{q}_j \chi(r_{ij}) + E_{D3} \quad (18)$$

where E_i is the predicted energy contribution for atom i , $\hat{q}$ is the predicted partial charge, χ is a smoothed approximation of Coulomb interaction r_{ij}^{-1} , and D3 is the DFT-D3 dispersion correction.⁷¹ To evaluate its performance, PhysNet was applied to molecular dynamics simulations of an Ala₁₀ molecule. The optimized configurations and PES were shown to closely match the reference DFT level, demonstrating the accuracy and reliability of PhysNet.

2.4. Physics-Based MLFFs

The MLFFs mentioned above are capable of learning the PES of molecular systems based on atomic coordinates. Another strategy involves incorporating physical principles into MLFFs. One well-known example is Lennard-Jones interaction based FFs, which are often expressed as the short-range repulsion and long-range attraction, embedded as r^{-12} and $-r^{-6}$ terms, respectively. Other physical properties include bond orders⁷² and electronic densities.⁷³ Among these models, the MB-pol water FF calculates the energy of water molecules as the sum of many-body interactions:⁷⁴

$$E_N(1, \dots, N) = \sum_{i=1}^N \epsilon^{1B}(i) + \sum_{i<j}^N \epsilon^{2B}(i, j) + \sum_{i<j<k}^N \epsilon^{3B}(i, j, k) + \dots + \epsilon^{NB}(1, \dots, N)$$

where ϵ^{NB} represents N-body interaction term involving N atoms ($1, \dots, N$), expressed in the form of a permutationally invariant polynomial:

$$\epsilon^{NB} = \sum_l^L c_l \eta_l \quad (19)$$

where L represents the number of parameters, c_l are the linear parameters optimized using ML, and η_l are the corresponding polynomials. These coefficients could be fitted using ML techniques including but not limited to kernel ridge regression, Gaussian process regression, and artificial neural networks, by minimizing the differences between the PES predicted by MLFFs and those obtained from high-level quantum calculations. Paesani and co-workers have also developed tools MB-fit⁷⁵ and MBX⁷⁶ for the efficient training and application of data-driven MB-pol MLFF.

Another strategy is to directly solve the Schrödinger equation using ML. While the computational cost of solving it analytically is prohibitively high for MD simulations, Noé and co-workers have developed PauliNet, a deep neural network (DNN) framework based on variational quantum chemistry. Comparing to the traditional quantum Monte Carlo (QMC) method, DNN has higher flexibility in solving wave functions with reduced number of determinants.⁷⁷ In subsequent work, they have developed approaches to compute the excited-state solutions of Schrödinger equations within the QMC framework.⁷⁸

2.5. Long-Range Interactions in MLFFs

Most MLFFs discussed above learn atomwise energies based solely on local environments, which inherently neglects long-range effects, particularly electrostatics. Such charge–charge interactions are essential for accurately modeling aqueous solutions and become even more critical in systems with broken symmetry, such as charged interfaces.

One strategy to address this limitation is to predict explicit partial charges for each atom within the MLFF framework. Long-range electrostatics can then be computed using standard classical approaches, such as Ewald summation. Examples include embedding charge-prediction schemes such as Charge Equilibration (QEq), kernel QEq, machine-learning QEq models,⁷⁹ iterative charge equilibration,⁸⁰ and variational charge equilibration models.⁸¹ For example, in traditional Behler–Parrinello MLFFs, charge equilibration has been incorporated as an auxiliary neural network to predict atomic charges alongside short-range interactions, thereby enabling the treatment of charge interactions and charge transfer.⁸² Importantly, the “charges” used in these MLFFs are fictitious latent variables that reproduce the correct long-range energy. For instance, Cheng developed the Latent Ewald approach, in which each atom is assigned a hidden variable analogous to a partial charge but unconstrained by physical charge distributions; the model is trained such that the Ewald summation over these latent variables reproduces the true long-range interactions.⁸³ A complementary strategy avoids explicit charge assignment altogether by learning long-range interactions directly from local atomic environments. Ceriotti and co-workers introduced a long-distance equivariant representation that can be computed locally yet captures nonlocal electrostatic effects, outperforming local descriptors such as SOAP for charged systems.⁸⁴

2.6. Coarse-Grained MLFFs

Besides MLFFs designed to reproduce high-level quantum chemistry calculation at reduced computational cost, coarse grained MLFFs have been developed to accelerate large-scale molecular dynamics simulations, particularly for macromolecular and protein systems. A representative example is the CGnet model introduced by Clementi and co-workers, in which the loss function is defined as the L2 difference between the predicted and reference coarse-grained forces:

$$L(\theta) = \langle \|\xi(\mathbf{F}(\mathbf{r})) + \nabla U(\xi(\mathbf{r}); \theta)\|^2 \rangle \quad (20)$$

where ξ denotes the mapping from atomistic to coarse grained forces and θ represents the neural-network parameters.⁸⁵ The full CGnet architecture is illustrated in Figure 4. Structural features, such as pairwise distances, angles, and dihedral angles, are computed from the coarse-grained coordinates $\mathbf{x}$, and the invariances of the coarse-grained potential energy, which is strictly speaking now a free-energy, are preserved through a transformation $\mathbf{y}$. The total coarse-grained energy is expressed as

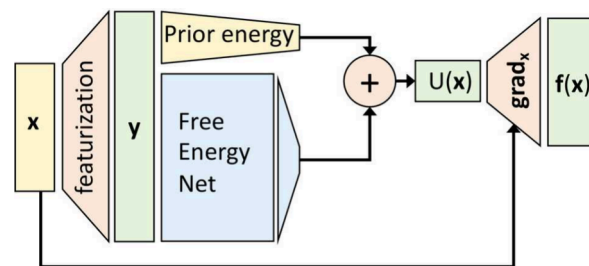


Figure 4. Architecture of the CGnet machine-learning coarse-grained force field. $\mathbf{x}$ denotes the coarse-grained coordinates, and $\mathbf{y}$ represents the transformation used to preserve the invariances of the free-energy surface. U and $\mathbf{f}$ are the predicted potential energy and forces, respectively. (Copyright 2019, American Chemical Society).

$U(\mathbf{x}) = U_0(\mathbf{x}) + U_{\text{Net}}(\mathbf{x})$, where U_0 is a physical regularizer enforcing appropriate configurational constraints, and U_{Net} is the neural network estimator. The resulting coarse-grained forces, $\mathbf{f}(\mathbf{x})$, are obtained as the gradient of the learned free-energy function U .

In subsequent work, the same group introduced CGSchNet, which replaces hand-crafted structural features with graph neural network representations.⁸⁶ Using these coarse grained MLFF models, simulations have successfully reproduced the native states of 12 proteins and mapped folding pathways across multiple metastable states.⁸⁷ More recently, the approach has also been shown to accurately capture the folding behavior of intrinsically disordered proteins.⁸⁸ While the same model did well for the 12 proteins included in the training set, transferability to other proteins different from these 12 proteins remains an exciting avenue for further research. Beyond biochemical systems, ML coarse-grained force fields have additionally been applied to model phase transitions in water.⁸⁹

2.7. Foundation Models of MLFFs

As discussed above, one goal in the development of MLFFs is to achieve broad compatibility and transferability, ultimately enabling the construction of universal force fields applicable across a wide range of MD simulations. In this context, using pretrained MLFFs, which provide general-purpose representations that can be efficiently fine-tuned for specific systems, substantially reduce the requirement of training data typically rely on computationally expensive quantum-chemical calculations.⁹⁰ Recently, several MLFF foundation models have been introduced, including CHGNet,⁹¹ MACE-MP, MACE-OFF,⁹² GRACE,⁹³ and M3GNet.⁹⁴ These models represent an emerging direction toward universal and adaptable force fields for molecules in MD simulations.

3. REACTION COORDINATE CONSTRUCTION FOR ENHANCED SAMPLING AND SIMULATION ANALYSIS WITH ML MODELS

3.1. Overview

Even with the rising popularity of MLFFs for computationally efficient and chemically accurate simulations, the issue of long time scale simulations still persists. As aqueous and interfacial systems are modeled by complex surfaces, the lack of accessible time scales limits the observation of relevant transitions and accurate thermodynamic/kinetic measurements. Hence, to fully explore the PES, the integration of enhanced sampling schemes with accurate force-fields becomes necessary.

More generally, irrespective of whether the simulation was enhanced or not, one significant shortcoming of MD simulations is their high-dimensional and noisy output, easily comprising terabytes of data for just one system. As quantifying physical phenomena of interest from raw trajectories can be regarded as very difficult, having a low-dimensional estimate of the reaction coordinate (RC) becomes necessary for easier interpretation. This dimensionality reduction makes it more manageable to identify patterns and features that might otherwise be obscured and has widespread use in methods such as metadynamics, umbrella sampling (US), and weighted ensemble. However, finding a suitable RC without *a priori* knowledge is quite challenging.

Oftentimes, an intuitive collective variable (CV) used to approximate the RC is not sufficient for sampling. While we can identify and attribute parameters such as pairwise distances and torsion angles to be of physical significance, they are not necessarily the best variables when it comes to biasing in enhanced sampling methods. Though they can successfully result in a better demarcation of different states in the FES, they are not naturally optimized for capturing the slow modes. Hence, the need to find fluctuating parameters sure to enhance a system's slow modes, which then helps with a faster convergence of the FES. In the case of nucleation, extensive studies have explored the construction of effective CVs.^{95,96} It is common to include a representation of the local bond order through Steinhardt order parameters which characterize the crystalline structure. However, while they are good indicators of the existence of different metastable crystalline like states, they do not always approximate the RC and are not apt at adequate representation of more glassy or polymorphic structures.^{97,98} This leads to either the addition of other CVs or the curation of one. When considering the clustering of NaCl ions, studies by Finney and Salvalaglio⁹⁹ as well as Giberti et al.¹⁰⁰ employ a CV that describes the local density of the system that does not predetermine a crystal structure. In lieu of this, machine learning based methods have been brought on to facilitate this process. One such approach being the projection of a high-dimensional trajectory landscape onto a regularized low-dimensional space meant to approximate the RC.

While there are numerous enhanced sampling schemes one could use, here we do not discuss them and instead refer to a review by Tiwary and van de Walle¹⁰¹ for an overview of the topic. We focus instead on the input features that are needed for these enhanced sampling methods to construct the RC or to analyze the high-dimensional data obtained from MD. This is an issue which is often ignored, even with ML based methods for learning the RC.¹⁰² While the protocol for constructing the RC might be based on ML, the assumptions made about how the input features that go into the ML model were constructed could have significant effects on the reliability of the subsequent enhanced sampling simulation or the analysis of the MD.

In this Section, we discuss different methods for learning the RC that use different flavors of ML, including methods that use ML for learning the RC from preselected features as well as more recent graph neural network based methods that automate the feature selection itself.

3.2. Dimensionality Reduction Techniques for Preselecting Collective Variables

Complex, high-dimensional trajectory data embodies the curse of dimensionality when it comes to featurization. In order to semiautomate the process of picking features without *a priori*

information, several dimensionality reduction techniques have been created with the objective of preserving key information. One of the earliest implementations of dimensionality reduction, principal component analysis (PCA) aims to linearly transform noisy trajectory data to a lower dimensional coordinate system which identifies important features. It does so by maximizing variance of the first components—eigenvectors of the covariance matrix. Its time scale counterpart, time-lagged independent component analysis (TICA) discussed in detail in Sec. 3.3, maximizes autocorrelations for a given time lag. Linear Discriminant Analysis (LDA), finds a linear combination of features which best separates data sets classes.¹⁰³ Though highly efficient, these methods are often linked by their use of linear transformations to handle complex data which limits a model's characterization of dynamics. This is quite significant when biasing dynamics as they may fail to fully explore the free energy surface. Recovering a higher resolution kinetic model would require the need for nonlinear transformations which has led to the application of the kernel trick to methods like LDA (KDA)¹⁰⁴ or TICA (ktICA)^{105,106} as well as methods that leverage the flexibility of neural networks like DeepLDA^{107,108} or Deep-TICA.¹⁰⁹ These models can be more expressive in terms of capturing dynamical features.

3.3. Time-Lagged Independent Component Analysis

The variational approach to conformational (VAC) dynamics¹¹⁰ provides a framework to estimate the slow modes of a system from eigenfunctions of the transfer operator. TICA¹⁰⁵ is one case of implementing VAC, widely used with Markov state models (MSMs).¹¹¹ In a Markov process, TICA attempts to find a maximally correlated projection vector for some time lag greater than zero. It does this by essentially solving the generalized eigenvalue problem:

$$C_{\tau}\mathbf{r} = \lambda_i \Sigma \mathbf{r} \quad (21)$$

where C_{τ} is the time-lagged correlation matrix and $\mathbf{r}$ is a projection vector with eigenvalue λ_i .

TICA is restricted to forming linear combinations of feature inputs to generate RCs which is optimal for schemes in biased metadynamics, but it limits the interpretability of the kinetic model. To address this, Schwantes and Pande¹⁰⁶ applied the kernel trick (ktICA) to create nonlinear RCs. By constructing a high-dimensional feature space, tICs can be nonlinear functions of the original coordinates. One drawback in the implementation of kernel methods is their comparatively slow computation. As a system of size N increases, the kernel calculation becomes intractable. It is important to note that when analyzing data, the calculation is not so impractical as opposed to when biasing dynamics. Further, it is also difficult to attribute physical meaning in kernel space.

3.4. VAMPnets

VAMPnets¹¹² implement a deep learning framework based on the variational approach for Markov processes (VAMP)¹¹³ to streamline and simplify the creation of MSMs from MD simulation data. It finds feature transformations χ_0 and χ_1 which optimize the VAMP score; typically the VAMP-2 score for its suitable application in time series data using the Koopman operator theory.¹¹⁴ As in eq 22, the VAMP-2 score R_2 is defined as the Frobenius norm of an $n \times n$ matrix $\mathbf{A}$, where C_{00} , C_{01} , C_{11} are covariance matrices. When the most significant Koopman singular functions (top m left and right) span the subspaces of respective features χ_0 and χ_1 , the VAMP-2 score is maximized.

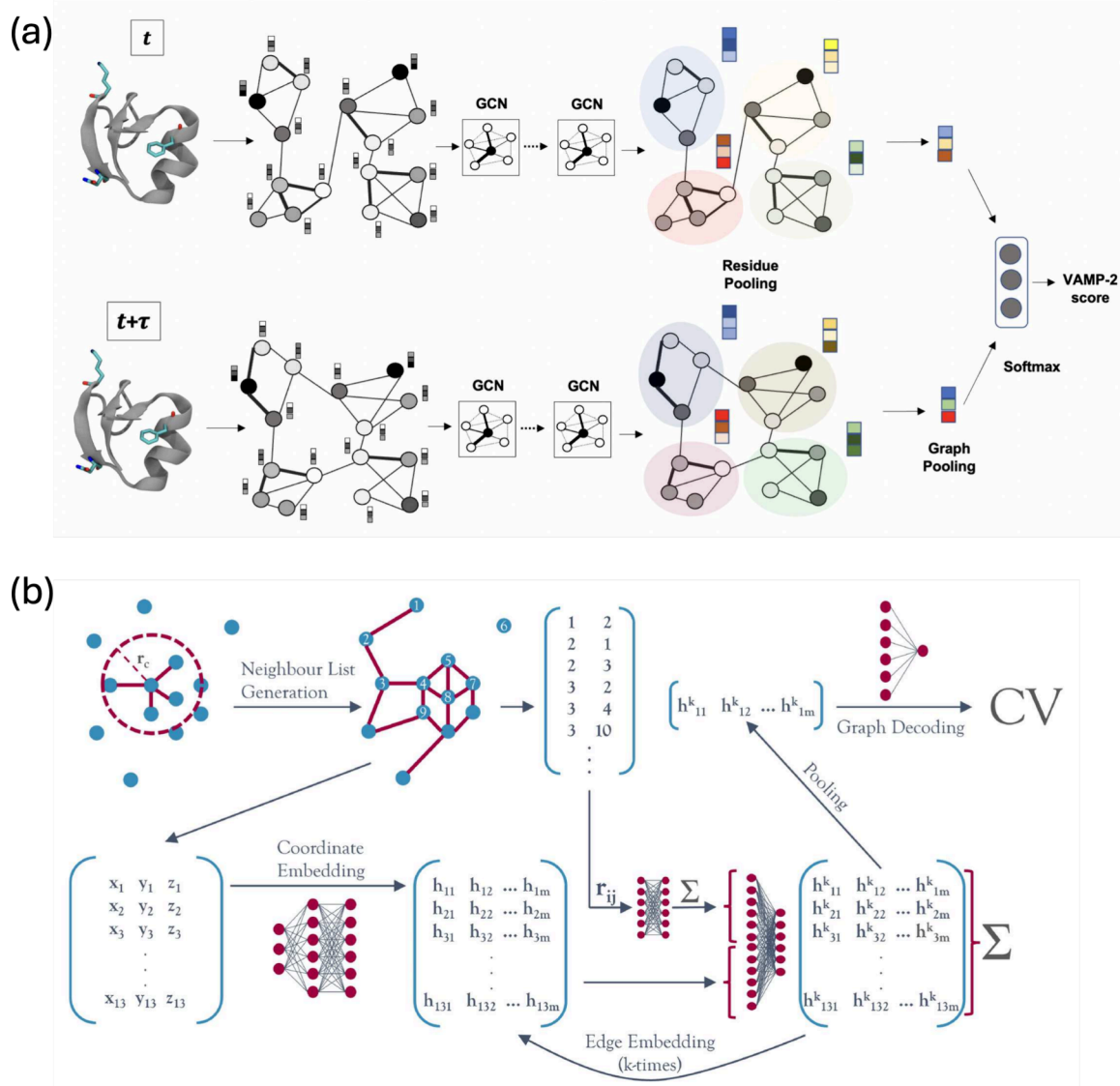


Figure 5. (a) GraphVAMPnet workflow schematic.¹²³ For a given time t and time-lag $t - \tau$ respective graph structures are created from molecular structures using a nearest neighbors scheme. Passing through graph convolution operations, hierarchical pooling is done to find the latent representation of the full graph which combines time t and $t - \tau$ for the optimization of the VAMP-2 score. (b) GNN model for learning nucleation CVs.¹²⁰ The graph structure is similarly created by a nearest neighbors algorithm. Cartesian coordinates are embedded via multilayer perceptron into a higher-dimensional representation which is updated by neighborhood information. A final pooling layer is used for final predictions. Panels (a) and (b) are reprinted from ref.¹²³ (Copyright 2022, AIP) and ref.¹²⁰ (Copyright, CC BY).

$$R_2[\chi_0, \chi_1] = \|C_{00}^{-1/2} C_{01} C_{11}^{-1/2}\|_F^2 \quad (22)$$

Optimal features χ_0 and χ_1 are thus able to capture the slow dynamics of the system. Similar to TICA, VAMPnets utilize the transfer operator to rank eigenfunctions and identify the relevant modes of a system.

The architecture of VAMPnets typically consists of two identical network lobes that perform dimension reduction, discretization, and featurization employing fully connected networks with a softmax output layer. These respective network lobes are trained to learn features χ_0 and χ_1 . As the softmax layer provides a representation of the discretized metastable states, it is somewhat unsuitable for enhanced sampling. Though the primary purpose of VAMPnets is that of state clustering to analyze kinetics, Ferguson et al. posited state-free reversible VAMPnets (SRVs)¹¹⁵ with linear and nonlinear output layers. It

follows the transfer operator method as its primary objective without the need of a kernel trick, in contrast to TICA.

3.5. State Predictive Information Bottleneck

By capturing relevant or meaningful information from compressed data, Information Bottleneck (IB)¹¹⁶ based methods maximize the trade-off between a model's accuracy and complexity. The resulting regularized low-dimensional representation can then be used to approximate an RC. Here we focus on State Predictive Information Bottleneck (SPIB).¹¹⁷ SPIB attempts to discretize the metastable states of a system given a time delay, Δt , alongside optimizing the RC. To improve latent space regularization, it uses a VampPrior¹¹³ mixture of Gaussians for the prior distribution. The RC is approximated by the weighted combination of CVs which are updated on-the-fly during training. The user has the freedom to choose either a linear or nonlinear encoder. The objective function optimized:

$$\mathcal{L}_{IB} = I(z, y) - \beta I(X, z) \quad (23)$$

comes from an updated version of RAVE¹¹⁸ which uses the difference between two mutual informations. In eq 23, $I(z, y)$ is the prediction capacity denoted by the mutual information between latent representation z and target prediction y . The second term, $\beta I(X, z)$, is the model complexity as defined by the mutual information between the latent and input X .

3.6. Graph-Based Reaction-Coordinate Construction

As an alternative to the difficult task of finding physically significant CVs to completely and accurately describe a system, graph neural networks (GNNs) have emerged as a solution. In place of having hand-crafted CVs, GNNs have been used to automate their selection, and even further, bypassing the need for them.^{119,120} Besides this advantage, they have also been growing in popularity for their intuitive representation of atomistic or molecular systems.

A collection of $\mathcal{V}$ nodes and $\mathcal{E}$ edges, a graph is typically denoted by $\mathcal{G}(\mathcal{V}, \mathcal{E})$. Graphs have three basic components: node features, edge features, and edge indices. Node features encode per atom, or per unit, characteristics such as species, mass, charge, or other relevant attributes. In the simplest case, a single atom is assigned to a node. Edge features describe the connection between nodes by using, for example, a Euclidean distance measure. Edge indices, or an adjacency matrix, denote which nodes are connected. A notable feature of GNNs is their use of message passing, which updates the model based on the exchange of information between nodes. By design, their architecture lends to the preservation of symmetries through translation and rotation transformations, which can be made to be invariant⁶⁸ or equivariant.¹²¹

As atomic coordinates can be taken as an input, methods have leveraged the flexible framework that GNNs offer to extract a system's slow modes.¹²² Figure 5 shows some workflows that follow this objective. Figure 5 (a) is the schematic used in GraphVAMPnets where a system's dimensionality is reduced to optimize the VAMP-2 score in order to uncover slow modes.^{122,123} Figure 5 (b) shows the schematic for learning nucleation CVs¹²⁰ via the implementation of multilayer perceptrons.

Given the large number of graph based methods that have been proposed, here we go through some such key approaches:

3.6.1. GraphVAMPnet. A modification of VAMPnets to include graph representations, GraphVAMPnets¹²³ efficiently learn the dynamics of a system concurrently lending to a more interpretable Markov model. The creation of a protein graph representation considers M nearest neighbors for a protein's C_α atoms. The edge embedding is denoted by adjacent node distances seen in eq 24

$$e_{ij}^t = \exp\left(\frac{-(d_{ij} - \mu_t)^2}{\sigma^2}\right) \quad (24)$$

where e_{ij} are Gaussian distances and d_{ij} is the distance between atoms i, j . The node embeddings are subsequently updated with the following:

$$v_i^{k+1} = v_i^k + \sum_{j \in N_i} w_{i,j}^k \odot g(z_{i,j}^k W_c^k + b_c^k) \quad (25)$$

where i represents a node at layer k ; W_c and b_c are convolution weights and biases, respectively; $z_{i,j}^k$ is the concatenation of nodes i, j at layer k along with edge embeddings e_{ij} ; $w_{i,j}^k$ is the sigmoid

function; and $g()$ is the edge-gating mechanism.¹²⁴ A graph pooling layer is done to capture the whole graph embedding.

To learn the feature representation of nodes for protein-folding dynamics, SchNet,⁶⁹ a continuous-filter CNN, is used. Edge attributes are thus calculated by a radial basis function (eq 26) that is then fed into a filter-generating function that maps these attributes to a corresponding filter.

$$e_{ij} = \exp(-\gamma(d_{ij} - \mu_t)^2) \quad (26)$$

When applied to node embeddings, this filter allows SchNet interaction blocks to update the many-body interactions between every atom j within a neighborhood of atom i . With an additional attention layer, the embeddings are updated with respect to the importance given to edge embeddings. As with VAMPnets the maximization of the VAMP-2 score is used for validation. The reconstruction of the free energy landscape shows clear separation of metastable states.

3.6.2. Graph Neural Network - State Predictive Information Bottleneck. Modifying SPIB to include graph-based inputs, GNN-SPIB takes advantage of graph representation to eliminate the need for hand selected feature inputs in favor of a fully automated process. Given an input trajectory, each time frame is defined as a geometric representation of a graph. Hence, the SPIB loss function is modified as follows:

$$\mathcal{L} = I(z, S_{t_n+\Delta t}) - \beta I(G_{t_n}, z) \quad (27)$$

where G denotes input graph at time t_n and $S_{t_n+\Delta t}$ is analogous to target y in eq 23. In preserving the permutation-invariant properties of GNNs, the inclusion of invariant input features as well as invariant operations within message passing is adhered to. Given this property, the graph message passing layers have flexibility when it comes to choosing GNN architectures. As of note, some schemes may be more computationally efficient than others, with consideration for enhanced sampling methods.

3.6.3. geom2vec. Drawing inspiration from natural language processing's (NLP) word2vec model, geom2vec¹²⁵ aims to map Cartesian coordinates to vector representations via pretrained GNNs. Additionally, it uses a denoising objective on atomic coordinates to improve performance. To ensure the vector representations transform accordingly for a physical system, the ViSNet¹²⁶ architecture, an equivariant geometric graph transformer, is used. The last message-passing layer activation parameters are used as input for later tasks, when pretraining on this architecture. The protocol for which is adding displacements from a multivariate Gaussian to the systems' coordinates to be learned, akin to learning a force field. As node-wise graphs are high dimensional, post-training calculations can be performed on a subset of atoms through coarse-graining operations. Given a learned representation $\mathbf{h}_i$ for atom i , a coarse-grained representation $\bar{\mathbf{h}}_{S_m}$, defined by eq 28, denotes average geometric information, termed "token." The term S_m denotes a subset of atoms belonging to a token. A trained geom2vec model can then be evaluated with VAMP or SPIB frameworks as done by Pengmei et al. to verify the dynamics of a system.¹²⁵

$$\bar{\mathbf{h}}_{S_m} = \sum_{i \in S_m} \mathbf{h}_i \quad (28)$$

3.6.4. Graph Based Nucleation Collective Variables. Even before deep GNNs, ideas involving graphs in one form or the other have been widely used to enhance and analyze

nucleation phenomena. Using a colloidal system, Dietrich et al.¹²⁰ construct a graph based model to approximate CVs for nucleation. Given that nucleation is a complex mechanism to observe, these CVs often include coordination numbers and sixth-order Steinhardt parameters ($n(Q6)$). As Steinhardt order parameters are based on spherical harmonics, of which linear combinations are rotationally invariant, they satisfy the justification for graph representation. Shown by eq 29, graph-level CV predictor ϕ pools nodes $\mathbf{x}_i$ through permutationally invariant pooling as denoted by $\oplus$.

$$f(\mathbf{X}) = \phi(\oplus \psi(\mathbf{x}_i)) \quad (29)$$

To ensure the connections of the graph for global and node-level predictions, a similar pooling operation is done on individual neighbors:

$$f(\mathbf{X}) = \phi\left(\oplus \psi_N\left[\mathbf{x}_i, \sum_{j \in \mathcal{N}_i} c_{ij} \psi_E(\mathbf{r}_{ij})\right]\right) = \phi(\oplus \mathbf{h}_i) \quad (30)$$

where E and N note edge and node predictions, respectively. ψ is a multilayer perceptron which maps to the node vector (ψ_N), or makes an edge-level prediction (ψ_E). c_{ij} are edge weights, assumed constant and set to 1. $\mathcal{N}_i$ is the neighborhood of node i . This is analogous to defining internal representation $\mathbf{h}_i$ which contains node information about its nearest neighbors. Modifying eq 30 to account for CV $n(Q6)$:

$$f(\mathbf{X}) = \sum_i \phi(\psi_N[\mathbf{x}_i, (\mathbf{x}_j | j \in \mathcal{N}_i)]) \quad (31)$$

where $\psi_N = \sum_{j \in \mathcal{N}_i} \sum_{m=-6}^6 q_{6m}^*(i) q_{6m}(j)$ and $\phi(\psi_N) = 1$ if ψ_N is above a certain threshold. Being that the model prediction lacks $E(3)$ -invariance, it would need to learn the effect of proper transformations on the CV. As such, calculating vector $\mathbf{r}_{ij} = \mathbf{x}_j - \mathbf{x}_i$ for edge-level, ϕ_E , predictions in reduced coordinate space would make the computation more manageable.

4. APPLICATIONS OF ML-BASED MD SIMULATIONS TO BULK AND INTERFACIAL AQUEOUS SYSTEMS

4.1. Structure and Dynamics of Aqueous Solutions

4.1.1. Bulk Water. Although MD simulations with MLFFs have gained popularity in recent years, MLFFs for bulk water have been under development for quite some time. In 2002, Cho et al. introduced a back-propagation of error algorithm to enhance the TIP4P model, making it polarizable and improving its ability to accurately reproduce potential energy in agreement with experimental observations.¹²⁷ Recently, by training with path-integral DFT-MD simulations, the DeePMD MLFF reproduced the radial and angle distribution functions of bulk water as well as other small organic molecules comparing to DFT simulations with the SCAN functional.⁵¹ Another feature of liquid water that had not received sufficient attention in simulations was its dielectric response. Another critical property of water is its multipole moments, which strongly influence the behavior of charged species in aqueous environments. Although fixed-charge classical MD can compute these quantities, the lack of polarization leads to inaccuracies; therefore, calculations with DFT-MD-level accuracy are preferred. The DeePMD framework can learn electronic structure using maximally localized Wannier functions (MLWF) from MD simulations, enabling it to predict the dielectric constant of bulk water.^{128,129} Calculating the dielectric constant in MD simulations with periodic

boundary conditions requires simulations on the nanosecond time scale to achieve convergence of dipole fluctuations.^{130–133} Using DeePMD, MD simulations with 512 water molecules over nanosecond time scales can be performed efficiently, accurately reproducing the multipole moments of water in agreement with DFT-MD results. These simulations have been conducted across various geometries, including single water molecules, bulk water, and a water cavity with a radius of 4 Å.¹³⁴ For simple systems such as bulk water, training MLFFs using high-level quantum data, such as the gold-standard CCSD(T) method, is feasible and can substantially improve predictions of the peak heights of radial distribution functions.¹³⁵ However, classical force fields often model water without polarization or rely on empirical polarization, which lacks the rigor of “first-principles” methods. Consequently, dielectric constants calculated from classical MD simulations are highly dependent on the selected, predefined water model, highlighting the advantage of MLFFs in bridging this gap.

4.1.2. Ion Solution. Modeling aqueous solutions in the presence of solutes presents a greater challenge for MD simulations with MLFFs compared to bulk water. Solutes including both simple and molecular ions are often charged, requiring MLFFs to accurately determine their partial charges even for simplest ions, especially when modeling high-valent ions of transition metal ions due to their strong polarization effects and complex atomic orbital configurations. In classical FFs, one simple solution is to scale the charges fractionally; however, the optimal scaling coefficients are not easy to determine in advance. Using the DeePMD framework, Lu and co-workers studied molten mixtures of MgCl_2 , NaCl , KCl , and trace amounts of water. The results demonstrated that MD simulations with MLFFs achieve accuracy comparable to DFT-MD in predicting thermodynamic properties such as density, diffusion, and viscosity.¹³⁶ Chen and co-workers used MLFFs to investigate the structure of $\text{LaCl}_3\text{--H}_2\text{O}$ solutions at various temperatures.¹³⁷ Similarly, MLFFs based on DeePMD have successfully reproduced MD simulations of other high-valent metal ions, including Zn^{2+} ,¹³⁸ Mn^{2+} ,¹³⁹ Ca^{2+} , and Mg^{2+} ,^{140,141} at the DFT level accuracy.

Beyond calculating solvated structure of ions, simulating the dynamics of electrolytes in aqueous solutions is both challenging and intriguing. Achieving converged dynamical properties requires large simulation boxes and sufficiently long simulation times, which are difficult to accomplish with DFT-MD simulations. The limitation makes MLFFs well-suited for studying such systems. Typical dynamical properties that have been calculated using MLFFs that would have otherwise been inaccessible at the DFT level, including the diffusion and rotational motion of water and ions, hydrogen bond lifetimes, as well as macroscopic properties such as liquid viscosity and dielectric response.¹⁴² Balasubramanian and co-workers conducted MD simulations of NaCl and CsCl solutions using the DeePMD MLFF, trained with DFT simulations from the revised PBE exchange-correlation functional incorporating D3 dispersion corrections.¹⁴³ CsI solutions accelerate water diffusion, whereas NaCl solutions reduce it. However, nonpolarizable force fields predict slow water dynamics around both ions, reflecting their inability to resolve subtle ion-dependent hydration effects. DeePMD has also been employed to simulate NaCl solutions, revealing the mechanism by which salt solutions reduce the dielectric constant as salt concentration increases.^{144,145}

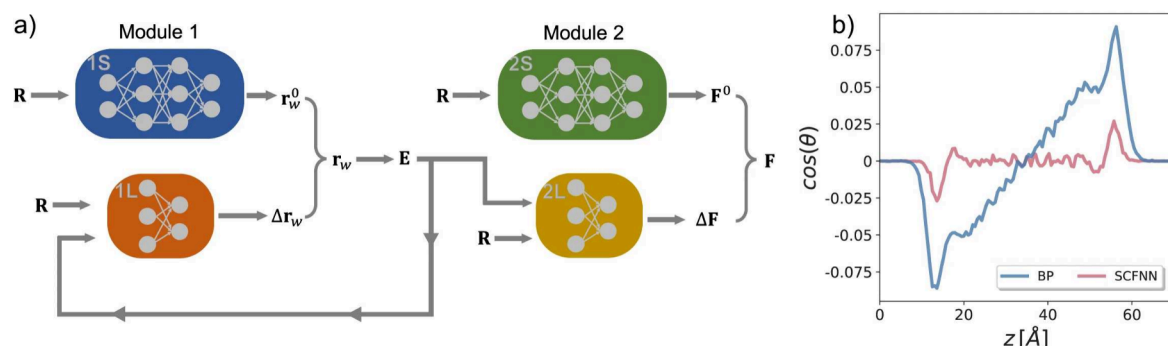


Figure 6. Framework of self-consistent field neural network and its result on MD simulations of the water/air interface. (a) The modules of SCFNN that take atom coordinates (R) as input and predict maximally localized Wannier function centers (r_w) and forces (F). The neural networks S and L learn the short-range contribution (r_w^0 and F^0) and long-range correction (Δr_w and ΔF), respectively. (b) Water orientation at water/air interfaces. z represents the distance from the interface. BP represents MD simulations using Behler-Parrinello MLFFs. The figure is reprinted from ref.¹⁵⁵ (Copyright, CC BY).

4.1.3. Interfacial Water. Another class of systems that benefits from MLFFs is water at interfaces, including planar interfaces, defected surfaces, nanoconfined environments, and water inside one-dimensional nanotubes.^{14,131,146–152} Generally speaking, aqueous interfaces play a critical role in materials science, particularly in electrochemistry, catalysis, and clean-energy applications. One property that differs significantly between bulk and interfacial water is the reduced dielectric constant. Although several hypotheses have been proposed to explain this behavior, a widely accepted one is that solid interfaces effectively truncate the bulk water environment; the absence of dielectric medium beyond the liquid suppresses the long-range dipole–dipole interactions, thereby reducing the overall dielectric response.¹⁵³ This effect is particularly pronounced at water/air interfaces, where the air side contains no molecules. MLFFs that rely on local atomic environments struggle to accurately reproduce certain properties, as learning long-range interactions is nontrivial. Limmer and co-workers demonstrated that MLFFs incorporating explicit electrostatic interactions achieve a good balance between training difficulty and accuracy.¹⁵⁴ Another promising solution is the development of the self-consistent field neural network (SCFNN),¹⁵⁵ which separates short- and long-range interactions to address the limitations of MLFFs that rely solely on short-range structure (Figure 6a). SCFNN can accurately reproduce the dielectric response of water under uniform external electric fields up to 3 V/nm. Water dielectric properties in these simulations are characterized by the water orientation parameter, $\langle \cos \theta(z) \rangle$, where $\theta(z)$ is the angle between the surface normal and the water dipole moment vector at a distance z from the surface, and $\langle \dots \rangle$ represents the average of all water molecules. This property is also used to estimate water vibrational spectroscopy signals observed in nonlinear optics experiments.¹⁶ At charged surfaces, interfacial water screens the electric field from the surface, resulting in $\langle \cos \theta(z) \rangle = 0$ in the bulk water region. In MD simulations with SCFNN, no preferred orientation is observed in bulk water, consistent with experimental observations. However, Behler-Parrinello MD simulations predict a long-range orientational order in bulk water, failing to capture the expected electric field screening (Figure 6b). This finding highlights that MLFFs based solely on short-range interactions cannot fully reproduce the effects of long-range interactions. MLFFs have also been applied to study the ice/air interface, which plays a critical role in ice melting and its impact on climate change. For instance, MLFFs have been used to investigate the

kinetics of ice growth.¹⁵⁶ When nuclear quantum effects (NQE) are included, the electronic bandgap of ice is observed to shrink, particularly in ice phases. This suggests that proton delocalization in ice significantly influences its physical and chemical behavior.¹⁵⁷

MLFFs have been increasingly applied to simulations of material surfaces to study the structure and dynamics of interfacial water. For instance, Selloni and co-workers performed MD simulations of water/ TiO_2 interfaces in the presence of small organic molecules, such as formic and acetic acids.¹⁵⁸ The simulations revealed how the water density and radial distribution functions are influenced by organic acid adsorption. The significance of TiO_2 surfaces in material science has led to several other MD simulation studies to explore their structure and properties.^{159–161} Studying the dynamics of interfacial water often requires long simulation times, which are difficult to achieve with DFT-MD simulations due to their computational cost. For instance, Yang and co-workers investigated water at various metal surfaces and observed superdiffusive rotation of water molecules near these surfaces, while their vibrational states and translational dynamics remained similar to bulk water.¹⁶² Beyond predicting structure and dynamics, MD simulations with MLFFs have also been employed to explore chemical and physical properties at interfaces. Examples include predicting the spin states of water/ LiMn_2O_4 interfaces,¹³⁹ molecular multipole moments of water,¹³⁴ Helmholtz capacitance,¹⁶³ and band alignment at aqueous interfaces.¹⁶⁴ One well-known phenomenon of interfacial water is the decrease in its dielectric constant. To investigate the origin of this behavior, Michaelides and co-workers observed an antiparallel alignment of water molecules and a reduced dipole moment of water within the first layer adjacent to solid surfaces.¹⁶⁵ These effects constrain the fluctuations of water orientation perpendicular to the solid surface, leading to the suppression of the dielectric constant. MLFFs are also applied to study water confined inside carbon nanotubes, which forms distinct one-dimensional water structure that differ from bulk water and water at planar interfaces. Pham and co-workers observed an enhanced dielectric constant of water along the axial direction within carbon nanotubes, which they attributed to changes in water dipole orientations and collective phases. This work highlights how MLFFs effectively combine the advantages of ab initio MD, such as accurate electronic properties, with classical MD simulations, which efficiently capture molecular configurations and dynamics.¹⁶⁶

4.1.4. Analysis of Aqueous Solutions Simulation Data Using ML Models. As discussed above, MD simulations generate the coordinates and velocities of each atom for each sampled configuration, with a dimension of $3N$, where N represents the number of atoms. To efficiently obtain physical insights from simulations, several order parameters, such as the radial distribution function, angular distribution function, tetrahedral order parameter, and hydrogen bond analysis, have been proposed as dimensionality reduction tools to describe water structure. In bulk water, each molecule donates and accepts an average of 3.5 hydrogen bonds, exceeding the threshold for a tetrahedral lattice and forming a connected hydrogen bond network.^{167,168} For interfacial water at planar water/solid interfaces, whether hydrophilic or hydrophobic, an enhanced connectivity of the hydrogen bond network is generally observed.^{169–172} For instance, the hydrogen bond network in a thin slab of bulk water is not connected, whereas it reaches connected at the Pt(111) surface (Figure 7a and b). At

solutions at high concentrations, the network of ion clusters exhibits the characteristics of random networks, where the degree distribution follows the Poisson distribution: $P(k) = e^{-\langle k \rangle} \frac{\langle k \rangle^k}{k!}$, with k and $\langle k \rangle$ representing the degree and the average degree, respectively (Figure 7c).¹⁷⁶ Furthermore, the graph spectra of the ion and water networks differ significantly as concentration increases, indicating a strong perturbation of the water network by the ions.^{175,177,178}

In addition to graph features that describe the global topology of water, local order parameters remain crucial for distinguishing the structural environment of water molecules. One widely used local descriptor is the tetrahedral order parameter, which characterizes the degree to which the hydrogen bonding environment of a water molecule resembles an ice-like or liquid-like structure.¹⁷⁹ The parameter has been found to strongly correlate with water autoionization.¹⁸⁰ Beyond human-intuitive descriptors, ML-based descriptors, particularly those derived from unsupervised learning, offer a distinct perspective for extracting meaningful structural information, with broad applicability and reasonable training cost.⁵⁰ For ML models, the SOAP descriptor is popular for distinguishing phase states of water and crystals.^{181–184} Integrating SOAP descriptors with intrinsic dimensionality estimation offers a robust framework for probing the structure and dynamic characteristics of ionic hydration shells. This combined approach also enables a more quantitative assessment of classical concepts such as structure maker and structure breaker articulated in the Hofmeister series.¹⁸⁵ SOAP has also been extensively used in constructing MLFFs, as discussed in Section 2.2. Recently, Pavan and co-workers developed timeSOAP, which employs time-series SOAP descriptors to reveal the dynamics of local water environments.¹⁸⁶ In subsequent work, they introduced an unsupervised onion clustering approach to identify fluctuations and dynamics in microscopic environments to distinguish ice–water interfaces in a fully data-driven manner.^{187–189} Time series trajectory has also been used to identify collective angular jumps of liquid water¹⁹⁰ the ferroelectric features¹⁹¹ and vibrational spectroscopy of supercool water.¹⁹² While SOAP is a generalized descriptor, it behaves differently from specific order parameters such as the orientational tetrahedral order, translational tetrahedral order, local structural index, distance to the fifth oxygen, coordination number, number of hydrogen bonds, and Voronoi density. Notably, SOAP struggles to fully capture discrete structural features like hydrogen bonding, which are essential for understanding water's local structure, indicating that the trade-off between the generality and specificity of order parameters needs attention.¹⁹³ This highlights the ongoing need for developing specialized descriptors tailored to capture specific properties of water structure effectively.

Many complicated processes, for instance, ligand binding in biochemistry, can be studied using model simulations such as adsorption of small organic molecules at specific designed solid surface. Beyerle and co-workers designed simulations of a single Lennard-Jones (LJ) particle and C_{60} molecule adsorption to a spherical pocket, and found that the roles of entropy and enthalpy for the unbinding processes. For the LJ particle, unbinding is dominated by the change of particle solvation, whereas that for a larger C_{60} molecule needs the dewetting of the pocket. In addition, to design materials of certain hydrophilicity, Patel and Shell design ML models to predict the free energy of water wetting to a patterned surface containing hydrophilic and

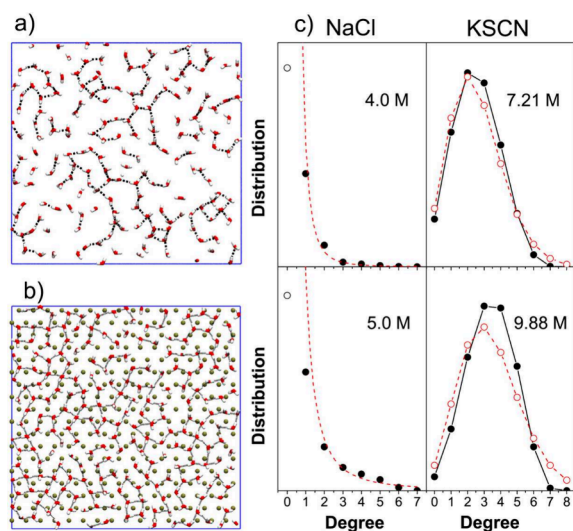


Figure 7. Water hydrogen bond structure in the perspective of graph theory. (a) and (b) Water hydrogen bond network in the bulk (a) and at the water/Pt(111) interface (b). Red, white, and yellow particles represent oxygen, hydrogen, and Pt atoms, respectively. Black lines represent hydrogen bonds. (c) Degree distribution of ions for NaCl and KSCN solutions. Black dots are calculated from MD simulations. Red dashed lines are estimated from theory: power law for NaCl and Poisson distribution for KSCN. Panels (a) and (b) are reprinted from ref.¹⁷³ (Copyright 2024, APS), and panel (c) is reprinted from ref.¹⁷⁶ (Copyright 2015, AIP).

some interfaces, such as air, alumina, and Pt(111), the number of water molecules required to cover the two-dimensional space is lower than that in a bulk water slab.¹⁷³ Graph-based models have also been applied to analyze ion solutions. For salts with high solubility, such as KSCN, ions tend to form an extended ion network that intertwines with the water hydrogen bond network. In contrast, salts with low solubility, such as NaCl, exhibit a phase separation between the ion network and the water network.^{174,175} Using graph properties as structural descriptors, such as the degree distribution and graph spectrum, NaCl is observed to exhibit characteristics of a scale-free graph, where the degree distribution follows a power law: $P(k) \propto k^{-\gamma}$.¹⁷⁶ Increasing the concentration of NaCl causes minimal perturbation to the graph spectrum of both the ion network and the water hydrogen bond network. In contrast, for KSCN

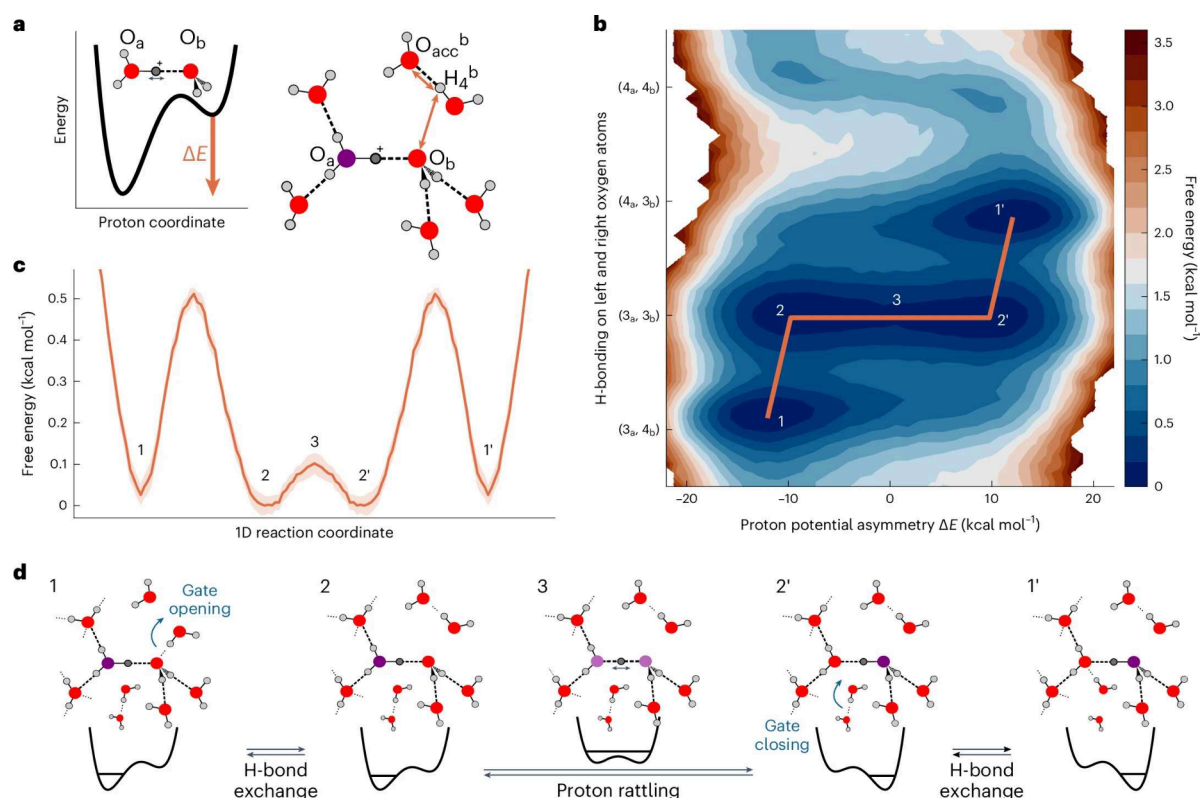


Figure 8. Free energy surface for proton transfer. (a) Definition of collective variables: the proton potential asymmetry ΔE and the number of hydrogen bonds. (b) The two-dimensional FES. (c) The 1D FES obtained using the yellow line in plot (b) as the reaction coordinate. (d) Snapshots of the proton transfer process, corresponding to states 1, 1', 2, 2', and 3 marked in plot (b). The figures are reprinted from ref.²⁰⁷ (Copyright 2024, Springer Nature).

hydrophobic units from certain chemical function group.^{194,195} Unlike ordered surfaces widely used in modeling inorganic materials, these surfaces are extremely heterogeneous, and machine learning can do a good job to identify these patterns.

4.2. Proton Transfer

4.2.1. Proton Transfer in Bulk Solutions. The simplest chemical reaction in bulk water is water autoionization, a special case of proton transfer and simulations of the system have a long history.^{196,196–201} In pure water, the concentration of $[H^+] = [OH^-] = 10^{-7}$ mol/L, corresponding to an energy barrier of approximately 40 kJ/mol.¹⁸⁰ While the choice of DFT functionals for such systems is a longstanding debate, an even more important but often overlooked limitation is the sampling. Typically, DFT-MD simulations are performed on time scales of a few picoseconds and with simulation boxes containing around one hundred water molecules. As a result, the finite size effect that leads to artifacts in the structure of the outer hydration shell of proton or hydroxide ions undermines the accuracy of proton transfer calculations in DFT-MD, due to computational constraints, because accurately recovering bulk-like behavior requires simulation boxes several nanometers in length. In addition, overcoming the large energy barriers associated with proton transfer necessitates the use of carefully designed enhanced sampling techniques. However, the limited sampling in DFT-MD makes it impossible to achieve convergence for simulations involving complex CVs, resulting in errors in estimating proton transfer processes due to the loss of configurational details when using simplistic CVs such as the OH bond length or the OH coordination number at the DFT level.

Fortunately, MLFFs effectively address this issue. By training a deep neural network using the SCAN functional,²⁰² Calegari et al. calculated water autoionization in simulation boxes containing 64 to 1024 water molecules,²⁰³ and showed that the predicted free energy for water autoionization converges when simulations include 500 or more water molecules. The finite size effects primarily impact the separation of H_3O^+ and OH^- , with a minimum separation distance of approximately 0.8 nm required for accurate results. Although DFT-MD simulations with thousands of atoms are now feasible thanks to growing computational resources, MLFFs make such simulations significantly more accessible. Moreover, MLFFs enable simulations that span dozens of nanoseconds, far exceeding the time scales achievable with traditional DFT-MD. By analyzing the trajectories from these longer simulations, deeper physical insights into proton transfer mechanisms can be gained. For instance, Liu et al. uncovered an asynchronous and concerted proton transfer mechanism involving three protons, and identified a correlation between the intramolecular electric field and proton transfer processes.²⁰⁴

Proton transfer events are more common in acidic or alkaline solutions, where hydronium (H_3O^+) or hydroxide (OH^-) ions are introduced along with counterions.^{2,140} In these conditions, proton transfer occurs spontaneously, leading to the abnormally fast diffusion of H_3O^+ and OH^- in water via the Grotthuss mechanism.²⁰⁵ While hydronium in acidic solutions has been extensively studied, the diffusion mechanism of OH^- remains underexplored with machine learning methods, despite being significantly different from that of H_3O^+ .²⁰⁶ A particularly debated topic concerns the structure of the excess proton in

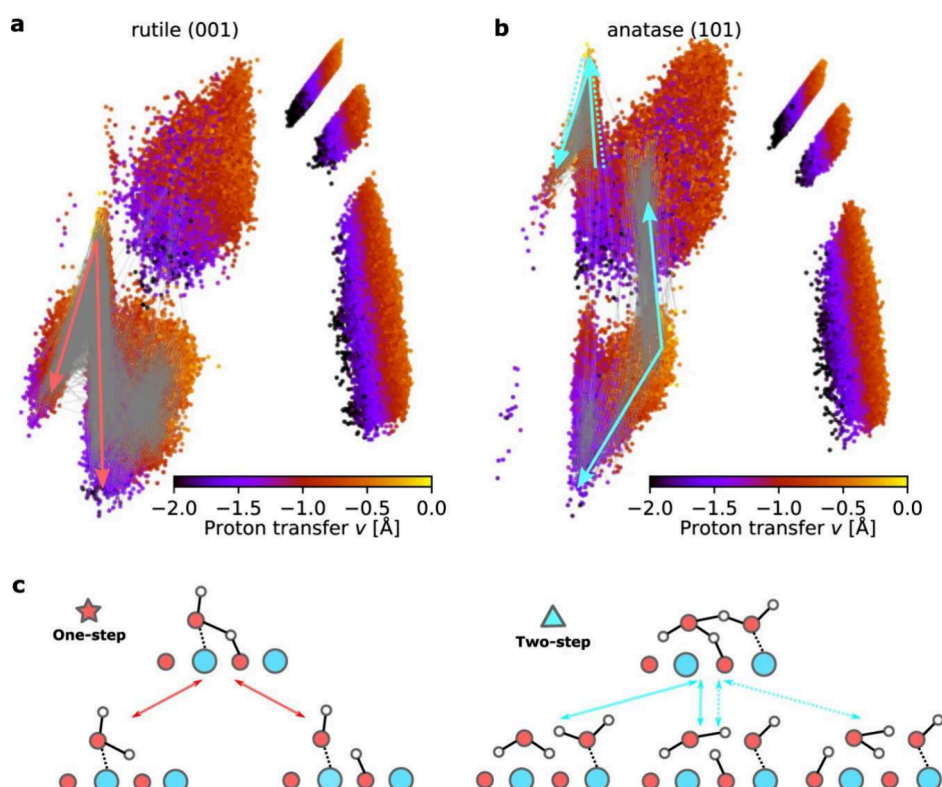


Figure 9. Mechanism of proton transfer processes analyzed using ML methods. (a) and (b) Kernel Principal Component Analysis of hydrogen environments in water/rutile(001) and water/anatase(101), respectively. The color scheme shows ν , the reaction coordinate based on the position of the proton. $\nu = 0$ represents the transition state, the proton is in the middle between two oxygen atoms. (c) Representation of one-step water dissociation (left) and two-step water dissociation-proton transfer (right, solid and dash arrows) mechanisms. Figures are reprinted from ref.²²⁴ (Copyright, CC BY).

hydronium. It can either belong to a single oxygen atom, forming an Eigen ion (H_3O^+), or be shared between two water oxygen atoms, resulting in a Zundel ion (H_5O_2^+). However, reaction coordinates based on OH distances often fail to capture a metastable state associated with the shared proton, conflicting with the two vibrational modes observed in 2D IR spectra.²⁰⁵ With the assistance of MLFFs, a new free energy surface (FES) for proton transfer in hydronium has been calculated, using the proton asymmetry potential energy and the number of hydrogen bonds of hydronium (represented by the oxygen–oxygen coordination number) as the collective variables (Figure 8b). These calculations show that the metastable state arises not from the location of the excess proton but from changes in the number of hydrogen bonds. For a traditional Eigen hydronium ion pair ($\text{H}_2\text{O}_a-\text{H}^+\cdots\text{H}_2\text{O}_b$), the oxygen atom (O_a) donates two hydrogen bonds, while O_b often accepts another hydrogen bond from solvent water. Since O_a does not form a complete tetrahedral hydrogen bond network, the excess proton resides closer to it. When one hydrogen bond donor leaves O_b , the proton binds more strongly to O_b , making it more shareable with O_b . This change results in a metastable state and a shift in vibrational frequencies. However, the excess proton does not lie at the center of the two oxygen atoms, indicating that the fully shared Zundel ion does not exist, consistent with the free energy calculation using proton location as the collective variable (Figure 8a, c, and d).²⁰⁷ Similar conclusions have been reached by analyzing the local environment of acidic solutions with unsupervised learning models. Hassanali and co-workers developed a workflow to study HCl solutions: calculating SOAP descriptors, estimating intrinsic dimensionality using

nearest-neighbor techniques, and applying the dimensionality reduction method Uniform manifold approximation and projection for dimension reduction (UMAP),²⁰⁸ and a single cluster is found for the hydronium structure, suggesting that Zundel and Eigen ions are thermodynamically indistinguishable, with their interconversion driven by fluctuations.²⁰⁸

While proton transfer is frequently discussed in the context of H_3O^+ in acidic solutions, the behavior of OH^- in alkaline solutions has received less attention, despite its distinct properties. Behler and co-workers conducted simulations of dilute and saturated NaOH solutions and identified concentration-dependent hydration exchange pathways. Specifically, a hydration water molecule can either leave the coordination shell of Na^+ directly, or proton transfer can occur before the resulting OH^- group dissociates.²⁰⁹ In subsequent work, the NQEs were incorporated into MLFFs using the ring polymer molecular dynamics model. While NQEs were found to have negligible effects on water- Na^+ interactions, they significantly lowered the energy barrier for proton transfer, highlighting the importance of quantum effects in accurately modeling NaOH dynamics.²¹⁰

4.2.2. Proton Transfer near Interfaces. Simulations of proton transfer events in bulk water provide valuable structural insights into these complex processes, especially when making use of ML force fields and models. From the perspective of energy materials design, proton transfer is particularly significant, especially when it is catalyzed at water/solid interfaces or under nanoconfinement. Simulating these systems requires modeling both the liquid and solid regions, with the liquid region being large enough to recover bulk-like behavior perpendicular and avoid finite size effects parallel to the solid

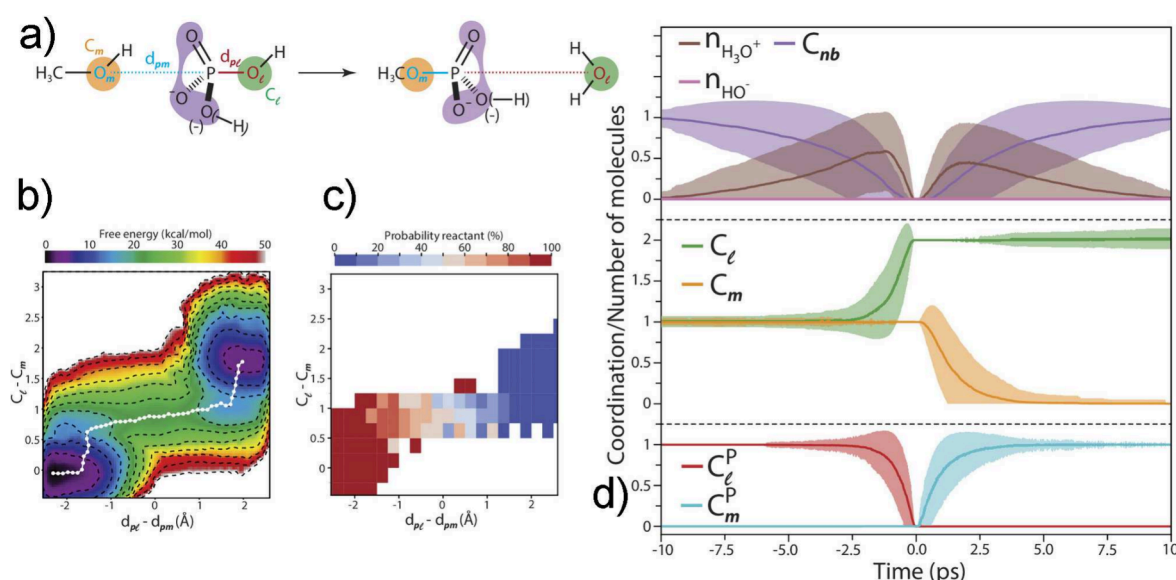


Figure 10. Simulation results of phosphoester bond formation reactions. (a) Schematic of the reaction between methanol and H_2PO_4^- . O_m and O_l represent methanol and leaving oxygen, respectively. C_m and C_l are OH coordination numbers with hydrogen of O_m and O_l . d_{pl} and d_{pm} are the distances of P and O_m , and P and O_l , respectively. (b) The free energy surface from OPES simulations with CVs: $d_{pl} - d_{pm}$ and $C_l - C_m$. (c) Committor analysis along OPES CVs. (d) Results of transition path sampling simulations. Additional order parameters: number of H_3O^+ ($n_{\text{H}_3\text{O}^+}$) and OH^- (n_{OH^-}), hydrogen coordination number with nonbridging carbon (C_{nb}), coordination number of $\text{P}-\text{O}_l$ (C_l^P), and $\text{P}-\text{O}_m$ (C_m^P). Figures are reprinted from ref.²²⁸ (Copyright, CC BY-NC-ND).

surface. This makes MLFFs well-suited for such simulations, as they can handle the required scale and complexity efficiently. Nanoconfined environments influence the reaction barrier and mechanism proton-transfer processes. For instance, Ogitsu and co-workers simulated proton transfer within hydrophilic TiO_2 slit pores and found that the reaction barrier is significantly reduced due to assistance from surface hydroxyl groups, underscoring the critical role of surface chemistry in governing reaction dynamics.²¹¹

Many energy materials are based on water/solid interfaces, where proton transfer is a key process.²¹² To simulate these systems, MLFFs have become increasingly popular as they significantly reduce computational costs and enhance configuration sampling compared to traditional AIMD simulations. Recent studies have explored proton transfer at various interfaces, including ZnO ,^{213,214} GaP ,²¹⁵ CoO ,¹⁶⁴ SnO_2 ,²¹⁶ CeO_2 ,²¹⁷ and TiO_2 .^{218,219} By employing MLFFs, the size of the solid supercell in these simulations can be extended to several nanometers, while simulation time scales reach the nanosecond range—far beyond the picosecond limitations of DFT-MD. For instance, Zhuang and co-workers investigated water dissociation at different layers of TiO_2 and observed odd–even oscillation effects in the properties of the solid layers. These simulations, made feasible by MLFFs, provided critical insights into the minimum number of layers required to accurately model water/ TiO_2 interfaces.²¹⁹

Oxide surfaces such as silica, alumina, and TiO_2 often feature hydroxyl (OH) groups when in contact with water. These surfaces can release or capture protons with water, resulting in charged surface. Charged surfaces have wide applications in electrochemical materials and play a crucial role in fuel cells and water splitting.²¹ The proton affinity to solid surfaces can be represented by the free energy change associated with proton transfer between the surface and water, which is computed using the free energy perturbation method.^{220–222} However, due to computational constraints, the simulation box size and the

number of water molecules are often limited. With the advent of MLFFs, these simulations have become significantly more accessible and provide more physical insights.²²³ Cheng and co-workers conducted MLFF-based simulations of water at various low-index TiO_2 surfaces commonly studied in material science, including anatase (100), anatase (110), rutile (001), and rutile (011).²²⁴ By employing kernel Principal Component Analysis (Figure 9a and b), they identified distinct proton transfer and water dissociation mechanisms: one-step, two-step, or both (Figure 9c). This combination of ML data analysis and MLFF simulations provides novel insights into proton transfer processes. Notably, configurations sampled using DFT-MD alone are often insufficient for comprehensive ML-based data analysis, further demonstrating the enhanced sampling power and utility of a combination of MLFFs and ML enhanced sampling.

The protonation states of solid surfaces have a significant impact on the behavior of the electrical double layer near the surface. Using a MLFF trained with DFT-MD simulations based on the SCAN functional, Zhang et al. investigated the anatase (101)-water interface.¹⁶⁰ Four independent systems were studied to examine the effects of ions and solution pH: (1) pure water, (2) NaCl solution, (3) NaCl with NaOH, and (4) NaCl with HCl. A key observation was the relatively larger EDL capacitance in a basic solution, highlighting the role of cation-surface interactions. This finding goes beyond the predictions of traditional EDL theory and aligns well with experimental observations.

Beyond material design, a longstanding and controversial issue in water science is the acidity of the water/air interface. Using MLFFs, both H_3O^+ and OH^- ions have been found to accumulate at the interface, albeit in different regions. This finding provides an explanation for the simultaneous observation of both ions at the surface.²²⁵ In addition to planar surfaces, MLFFs have been applied to study how OH^- accumulation promotes the nucleation of N_2 bubbles at the interface.²²⁶ These

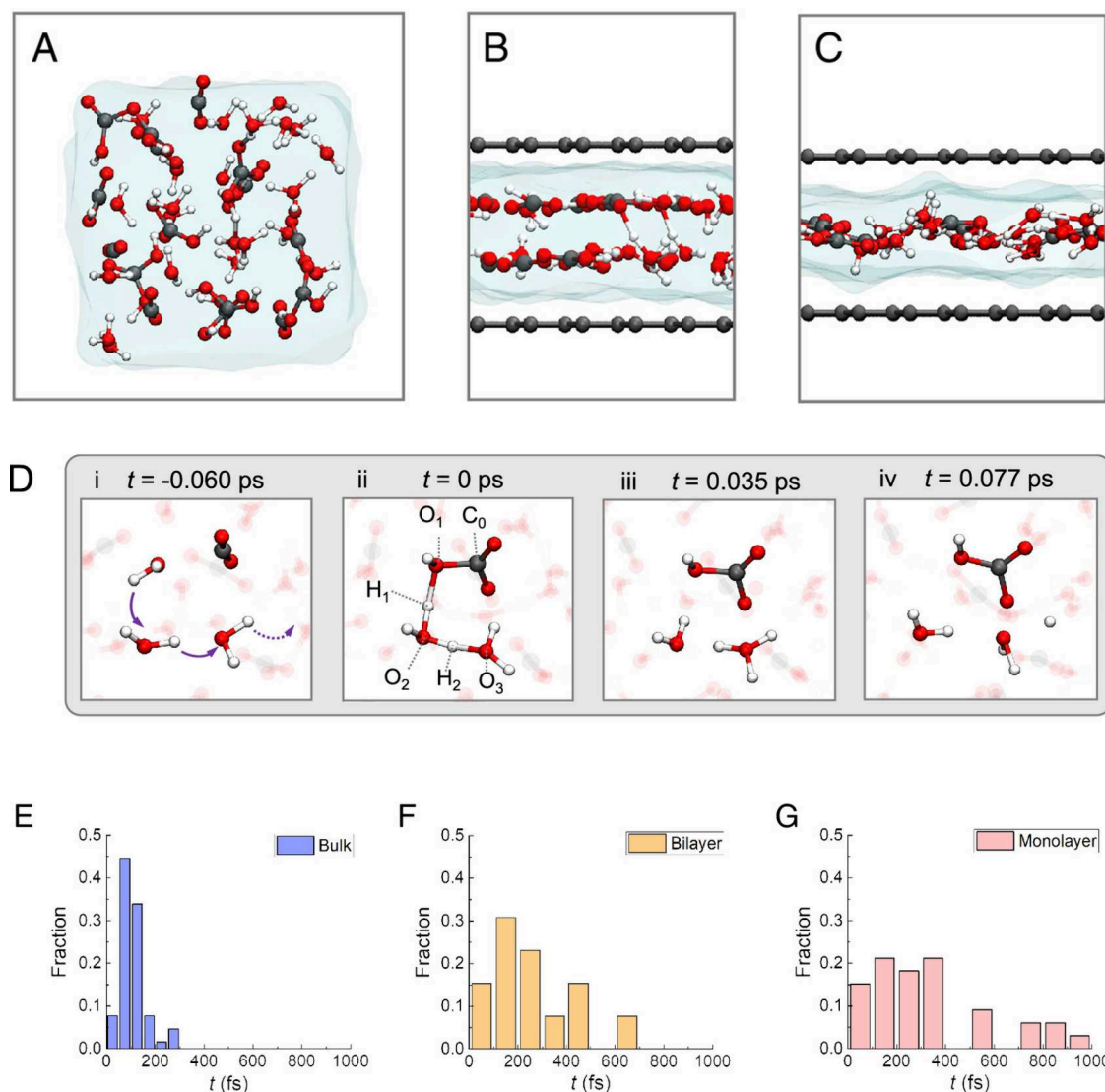


Figure 11. Simulation results of the dissolution of CO₂. (a–c) Schematic of designed MD simulations of bulk water (a), bilayer water (b), and monolayer (c). (d) Snapshots of the three-water proton transfer processes in bulk water. (e–f) Distribution of proton transfer time in simulations of the three environments. Figures are reprinted from ref.²³⁴ (Copyright, CC BY-NC-ND).

studies demonstrate the enhanced capabilities of MLFFs in facilitating MD simulations of complex systems, such as water interfaces, by providing insights that were previously challenging to achieve with traditional methods. Besides the water/air interface, simulations with MLFFs also decipher proton accumulation at the water/graphene interfaces.²²⁷

4.3. Catalysis

One key application of MD simulations is predicting the energy barriers and configurations of transition states in chemical reactions, traditionally dominated by DFT-MD. The use of MLFFs enables efficient sampling of diverse configurations during catalytic processes, significantly expanding the scope of such studies. In addition, the advancement of enhanced sampling methods, particularly the introduction of ML CVs, has greatly improved the ability to sample among desired states, enabling an accurate and comprehensive determination of free energy surfaces, which is essential for identifying transition state configurations and understanding catalytic processes. In this review, we will discuss catalysis in both biochemistry and

materials science. The former will focus on homogeneous catalysis, while the latter will address heterogeneous catalysis.

Catalysis in biochemistry, particularly reactions in aqueous solutions, is a key application area for MLFF simulations. One notable example is the formation of phosphoester bonds, a critical step in creating RNA oligonucleotides under prebiotic conditions. Previously, similar studies often relied on static energy calculations with implicit solvation models and primarily focused on the reverse process, phosphoester hydrolysis. With MLFFs and the on-the-fly probability enhanced sampling (OPES) enhanced sampling method, Stirnemann and co-workers performed microsecond-scale simulations with explicit water molecules.²²⁸ In the work, the CVs were chosen as the difference in P–O distances between the nucleophilic reagent and hydroxyl group, along with their coordination number differences for water hydrogen atoms (Figure 10a). Two-dimensional free energy surfaces were computed (Figure 10b), and these selected CVs were validated using transition path sampling with committer analysis (Figure 10c and d). The study required hundreds of nanoseconds of simulations, which is nearly impossible using traditional DFT-MD, especially with

high-level DFT methods. Another application of MLFF simulations in the origin of life involves peptide bond formation in aqueous solutions. Laage and co-workers used transition path sampling to identify two distinct reaction pathways: general base catalysis in neutral solutions and a direct cleavage mechanism in acidic and basic solutions.²²⁹ Many chemical reactions in aqueous solutions require the active participation of water as a solvent. For instance, the decomposition of urea, $(\text{NH}_2)_2\text{CO} \rightarrow \text{CNO}^- + \text{NH}_4^+$ and $\text{CNO}^- + \text{NH}_4^+ + \text{H}_2\text{O} \rightarrow 2\text{NH}_3 + \text{CO}_2$, involves water-mediated proton transfer. Explicit water models at the DFT level are essential for capturing these interactions accurately. Using well-tempered metadynamics, simulations have elucidated the role of water in facilitating proton transfer during these reactions.²³⁰

Heterogeneous catalysis plays a critical role in environmental and material science, and MD simulations provide valuable insights into these processes. Typically, environmental chemists are interested in the adsorption of molecules at water/air interfaces, which governs processes such as air pollution and carbon fixation. ML based MD simulations enable the investigation of complex reactions with reasonable computational cost.^{45,231,232} For instance, Limmer and co-workers studied the adsorption of N_2O_5 and found that its uptake occurs predominantly at the water/air interface rather than in the bulk phase. Using enhanced sampling MD simulations, they explored the interconversion of N_2O_5 , HNO_3 , and NO_3^- across the gas phase, liquid phase, and at the interface.¹⁷ Interfacial environments also influence the acidity of adsorbed ions, where the acidity of HNO_3 and HCOOH is weaker acidity due to partial desolvation. Their acidity recovers to bulk-like behavior only several angstroms away from the interface.²³³ Another example is the dissolution of CO_2 in aqueous solutions, both in bulk water (Figure 11a) and under nanoconfinement (Figure 11b and c). Using the Markov state model to analyze DFT-MD data (Figure 11d and e), Pan and co-workers found that in bulk water, the three-water proton transfer event occurred rapidly, in less than 200 fs, following a concerted mechanism. However, under nanoconfinement between two graphene surfaces, the same proton transfer takes longer, suggesting a stepwise mechanism.²³⁴ This work highlights the advantage of combining DFT-MD simulations with ML data analysis to study chemical reactions. We believe that simulations with MLFFs could generate larger data sets and further benefit from such approaches.

In materials science, designing and understanding electrochemical materials is a highly active research area. MD simulations can predict how the materials affect chemical reactions, such as the hydrogen evolution reaction (HER) and oxygen evolution reaction (OER) at solid interfaces.²³⁵ First, the structure of the aqueous solution often receives the most attention, while the contribution from surface properties is frequently overlooked. Behler and co-workers investigated the $\text{Li}_x\text{Mn}_2\text{O}_4$ surface to understand the oxygen evolution reaction (OER) in lithium-ion batteries.¹³⁹ Their findings revealed that water binds weakly to Mn(II) atoms, and that high oxygen coordination correlates with higher oxidation states of Mn atoms. Additionally, increased c_g electron mobility was observed at the surface, offering critical insights into the electronic and catalytic properties of $\text{Li}_x\text{Mn}_2\text{O}_4$ surfaces.

Furthermore, by combining MLFFs with enhanced sampling techniques, MD simulations can be used to calculate energy barriers during chemical reactions, thereby revealing more detailed structural information about catalytic mechanisms. For

instance, Liu and co-workers studied CO oxidation in both the gas phase and aqueous solution using GNN-based MLFFs combined with umbrella sampling. They calculated free energy barriers at various temperatures and decomposed entropy contributions into configurational, translational, and rotational components.²³⁶ Luber and co-workers combined MLFFs with ML-based collective variables to investigate the OER at the $\text{WO}_3(001)$ surface. Their simulations showed that products such as OH $^-$ and H_2O_2 are energetically favored over O_2 , providing valuable insights into the reaction pathways.⁵² Cheng and co-workers combined MLFFs with DFT-MD calculations to compute the redox potential of LiTFSI salt. This work uncovered a correlation between the water solvation structure and the electrochemical properties of the system, providing valuable insights for the design of advanced electrochemical materials.²³⁷ Parrinello and co-workers combined the DeepMD framework with the OPES method to investigate the effects of Li-imide surfaces on ammonia decomposition,²³⁸ and found out the role of dynamics, such as the flow of reactants, in catalysis, which is often overlooked in static calculations or DFT-MD simulations with insufficient time scales. This work highlights the power of advanced enhanced sampling methods in uncovering catalytic mechanisms and provides valuable insights for the rational design of catalytic materials, aiding experimental chemists in developing improved catalysts.

4.4. Phase Transitions with Emphasis on Nucleation

Phase transitions in condensed matter are inherently complex processes, involving multiple molecules at the microscopic scale and slow dynamics often spanning more than one length and time scale.^{239–242} These features make traditional experiments and MD simulations challenging tools for studying phase transitions. The involvement of large molecular ensembles over extended time scales necessitates large-scale and long-duration MD simulations, limiting the applicability of DFT-MD to only specific scenarios of phase transitions. Even for classical MD simulations, enhanced sampling techniques are invariably required to overcome the high reaction barriers associated with liquid–solid nucleation.⁹⁶ For simple molecules, biased MD simulations along specific CVs remain a popular approach for investigating nucleation. However, sampling along high-dimensional CVs is computationally impractical, and human-intuitive CVs may not guarantee optimal performance. To address these limitations, ML-based CVs have been developed (discussed in Section 3), offering a more effective framework for exploring complex phase transition mechanisms from high-dimensional configuration spaces.

While there are numerous types of phase transitions, here we start with an emphasis on nucleation processes. One would think that the nucleation phenomenon should be simple for the rare gas Argon, often modeled using Lennard-Jones particles.²⁴³ However, the nucleation rate of Argon deviates by several orders of magnitude from predictions made by theoretical frameworks, such as the classical nucleation theory (CNT).²⁴⁴ Traditional CVs for studying Argon nucleation include the mean and second moments of coordination numbers, which represent local density and density fluctuations. Tsai and co-workers applied the spectral gap optimization approach to refine nucleation CVs, quantifying the importance of density fluctuation changes during the nucleation process.²⁴⁵ Using infrequent metadynamics, they calculated the rate constant and found it to be more accurate when the second and third moments of coordination numbers were considered. In aqueous solutions, nucleation

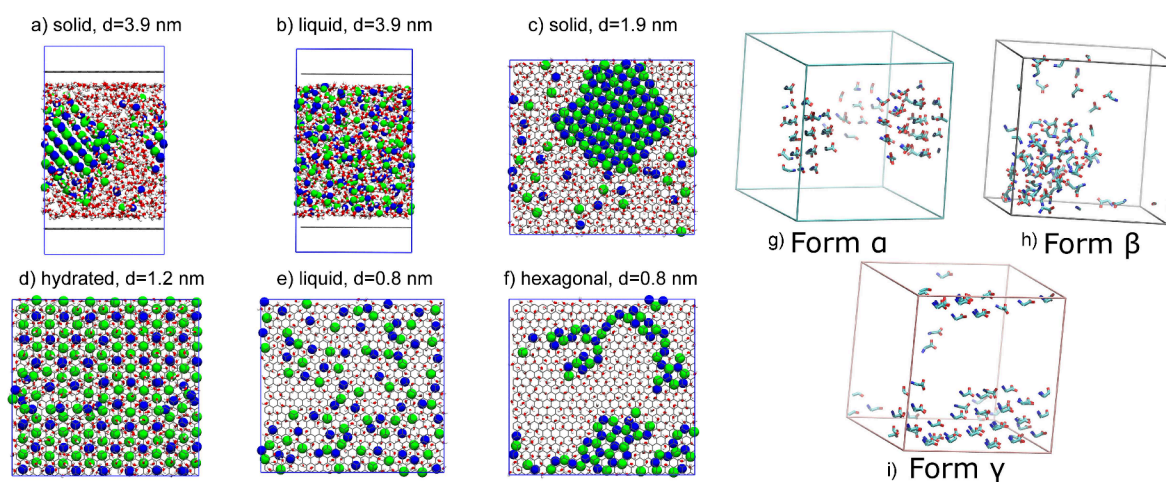


Figure 12. Phases sampled in ML enhanced sampling MD simulations. (a) to (f) are simulations of NaCl nucleation from nanoconfined aqueous solutions. d is the distance between two graphene sheets of the nanoconfinement. The snapshots are reprinted from ref.²⁴⁷ (Copyright, CC BY). (g) to (i) represent glycine phases: forms α , β , and γ . The snapshots are reprinted from ref.²⁵⁰ (Copyright, CC BY-NC-ND).

processes are even more complex. Wang et al. investigated NaCl nucleation and employed the SPIB method alongside interpretable ML models to analyze CVs from a set of order parameters describing local ion density, local ion order, and the number of solid-like ions.^{53,246} Their simulations revealed that ion local structure and solid-like ions, rather than ion density, drive the nucleation of NaCl from bulk aqueous solution or the melt. They proposed a mechanism where nucleation begins with an ordered assembly of ions in the liquid phase onto the surface of a solid precursor. Interestingly, under strong nanoconfinement, besides common face-centered-cubic solid (Figure 12a and c) and liquid (Figure 12b and e) phases, additional uncommon phases, including hydrated NaCl (Figure 12d) and hexagonal crystals (Figure 12f), were observed, consistent with experimental findings.²⁴⁷ Free energy calculations showed that nanoconfinement significantly enhanced the stability of solid phases. This effect is attributed to entropy suppression caused by restricted ion motion near graphene surfaces, analogous to a temperature decrease. To quantify the driving force of nucleation, the significance of various physical properties in simulations, represented by selected order parameters, was evaluated using the thermodynamically explainable representations of AI and other black-box paradigms (TERP) model.²⁴⁸ TERP analysis highlighted the role of hydration water, particularly the removal of hydration shells from Cl^- on the solid precursor, as critical to nucleation, challenging the traditional view that increased ion concentration drives liquid–solid nucleation. The dielectric properties of nanoconfined liquids were also identified to influence nucleation. The phase behavior of NaCl solution was found to be closely linked to that of a single Na–Cl pair. Using nonpolarizable classical force fields, nanoconfinement promoted the formation of Na–Cl pairs, aligning with the dominant solid phases seen in simulations. However, Michaelides and co-workers observed a contrasting trend with full MLFFs: a decrease in contact ion pairs and an increase in solvent-separated ion pairs, corresponding to hydrated phases.²⁴⁹ It is worth noting that the behavior of single ion pairs or low-concentration solutions may differ from that of high-concentration solutions, where solid phases could be stabilized under MLFFs. The disagreement between classical and MLFF-based simulations warrants further investigation. It is not surprising that the polarization of solid surfaces, often

neglected in classical force fields, significantly influences ion behavior near interfaces.

Predicting the phase transitions of multiatom molecules is more challenging than that of NaCl because nucleation processes must account for not only molecular density but also the relative orientations of the molecules.²⁵¹ Although these properties can be captured with human-intuitive descriptors, such as orientational entropy,⁹⁸ molecular order parameters⁹⁵ and local orientational similarity,^{252,253} Nonparametric ML descriptors provide a more general framework to distinguish crystal structures²⁵⁴ and to analyze dynamics during nucleation.²⁵⁵ These descriptors can also be incorporated into enhanced sampling MD simulations to drive reversible phase transitions, enabling systematic analysis of nucleation pathways. Zou and co-workers studied the nucleation of urea and glycine in bulk aqueous solutions using ML-based CVs, including average molecular orientation, pair orientation entropy, and interfacial water properties.²⁵⁰ These ML CVs enabled the simulation of nucleation processes and facilitated the back-and-forth phase transitions among different polymorphs of urea and glycine, especially several glycine phases including phase α (Figure 12g), β (Figure 12h), and γ (Figure 12i), demonstrating their utility in capturing the complex dynamics and structural transitions of multiatom molecular systems.

Investigating ice phases is a prominent topic, representing a longstanding question in physical chemistry.²⁵⁶ Historically, studies of ice structure have relied heavily on modified classical water models.^{240,257,258} While DFT-MD provides detailed insights, calculating phase stability with MLFFs is significantly more efficient, and an intriguing challenge is to evaluate whether the same MLFF can accurately reproduce the properties of both liquid and solid water at various pressures and temperatures.²⁵⁹ Car and co-workers employed DeePMD simulations trained on the SCAN functional, combining seeded simulations and OPES enhanced sampling with environmental similarity CVs.^{260,261} Their results successfully reproduced experimental nucleation rates and clarified how thermodynamic properties influence ice nucleation. The DeePMD MLFF, trained with the SCAN functional, has been applied to distinguish hexagonal (I_h) and cubic (I_c) ice structure. These simulations accurately reproduced experimental values for phase transition enthalpies and densities.²⁶² Furthermore, free energies of different phases were

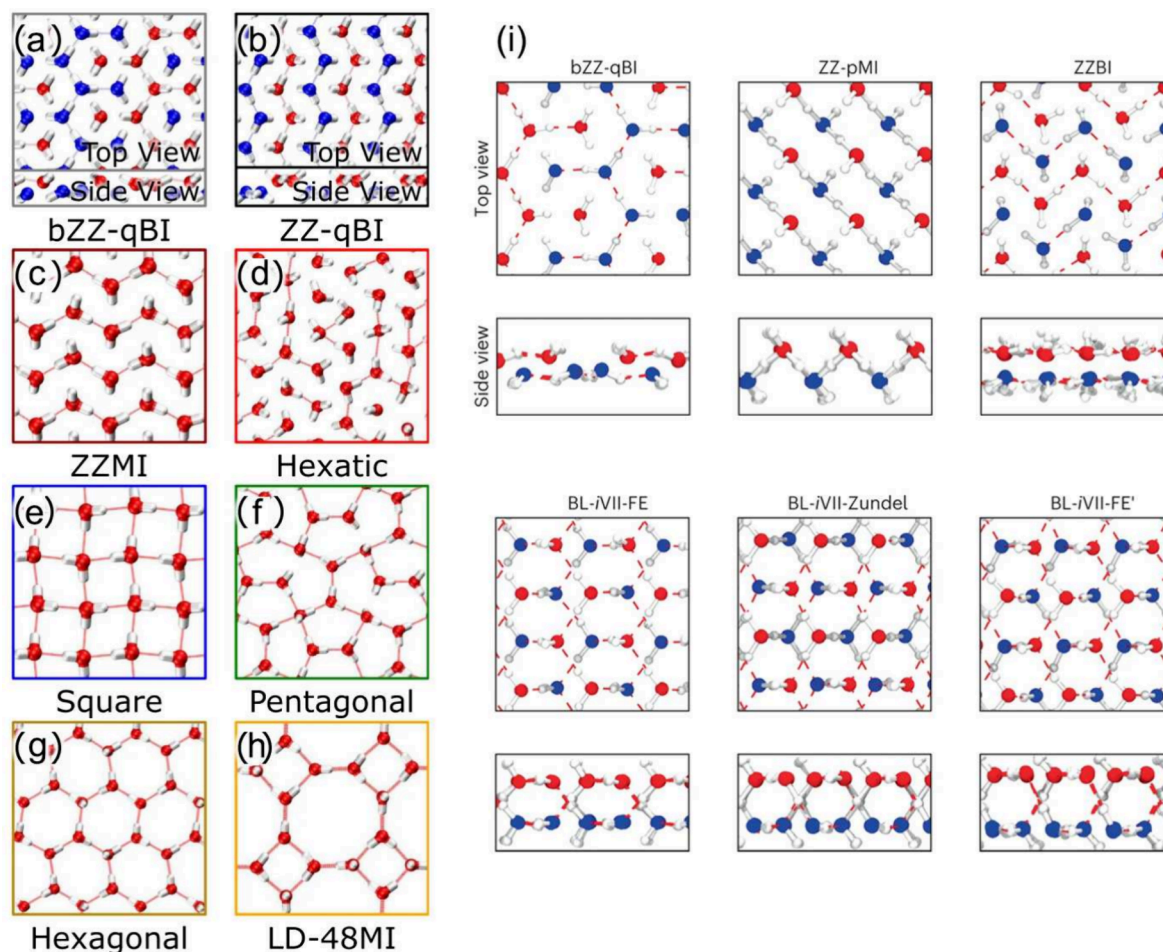


Figure 13. Snapshots of two-dimensional ice. (a–h) 2D ices under nanoconfinements. Beside common phases, the phases branched zigzag quasi-bilayer ice (bZZ-qBI), zigzag quasi-bilayer ice (ZZ-qBI), zigzag monolayer ice (ZZMI), and low-density 4-8² monolayer ice (LD-48MI). (i) Snapshots of top and side views of 2D protonated ice. pMI represents pucked monolayer ice, ZZMI represents zigzag monolayer ice, BL represents bilayer, iVII represents ice VII, and FE represents ferroelectric ice. Panels (a–h) are reprinted from ref.²⁶⁵ (Copyright, CC BY). Panel (i) is reprinted from ref.²⁶⁶ (Copyright 2024, Springer Nature).

calculated using reweighting algorithms for metadynamics, showcasing the combined power of MLFFs and enhanced sampling techniques in capturing the complexities of phase transitions.³⁹ Other MLFFs have also demonstrated their ability to capture the phase diagram of ice. For instance, the data-driven MB-pol model, trained from MD simulations calculated using the high-accuracy CCSD(T) method, reproduces a remarkably accurate phase diagram for water and ice. This includes liquid water and the ice *I*_h, II, III, V, and VI phases across temperatures of 200–300 K and pressures up to 1 GPa.^{33,263}

Ice formation near surfaces is a topic of significant interest, particularly for understanding environmental processes and climate change. Debenedetti and co-workers investigated ice nucleation near feldspar materials, focusing on the (100), (010), and (001) surfaces. Their simulations demonstrated the high accuracy of MLFFs in capturing ice nucleation processes. Using a simulation box containing approximately 30,000 atoms, they achieved a scale that is nearly impossible to study with DFT-MD.²⁶⁴ Another intriguing area of study involves ice under nanoconfinement, often referred to as “2D/monolayer ice”. MLFF simulations have advanced the understanding of these systems by reproducing various phases of the 2D ice (Figure 13a–h),²⁶⁵ the behavior of 2D ice with additional protons (Figure 13i),²⁶⁶ and the hydrogen bond network of 2D ice.²⁶⁷

These studies have significantly expanded our knowledge of uncommon ice phases and their properties under heterogeneous environments.

In addition to studying water and ice, water under high pressure is another growing research area for ML MD simulations.²⁶⁸ Debenedetti and co-workers tackled the long-standing problem of the liquid–liquid phase transition in water. By conducting simulations of liquid water across a range of temperatures and pressures, they performed a grid search and fitted the simulation data using a two-state model. Their findings pinpointed the liquid–liquid critical point $T_c = 224 \pm 3$ K, $P_c = 2,687 \pm 68$ bar, and $\rho_c = 1,072 \pm 16$ kg/m³.²⁶⁹ In subsequent work, they employed metadynamics to observe the back-and-forth transitions between high-density and low-density liquid states. These simulations allowed them to calculate the relative free energies of different states at specific temperatures and pressures, providing deeper insights into the liquid–liquid phase transition in water.²⁷⁰ Other observations have yielded different results. Paesani and co-workers identified a critical point at 198 K with a system size of 1,250 atoms using simulations with the MB-pol model.²⁷¹ Furthermore, Hassanali and co-workers calculated the FES using local and global SOAP descriptors combined with UMAP for dimensional reduction, and found that the FES exhibits one or two local minima, respectively. This

observation suggests that high- and low-density water cannot be distinguished solely based on their local solvation structure. Instead, identifying long-range contributions to water density variations is essential for understanding macroscopic density fluctuations.²⁷²

4.5. Prediction of Vibrational Spectra

The final application discussed in this review is the prediction of vibrational spectra using MD simulations. Accurate prediction of vibrational spectra requires not only the correct structural and dynamical representation from simulations but also sufficiently long simulation times to achieve converged spectra. From experimentalists' perspective, while vibrational spectra can be measured in the lab, the interpretation of the spectra often relies on other physical theoretical models, such as the EDL theory or MD simulations. Accurate simulated spectra enhance the reliability of the microscopic insights provided by MD simulations, making them more accessible and valuable for chemists seeking to understand the underlying chemistry.

In traditional classical MD simulations, the time scale of water vibrations is much shorter than the typical simulation time step, often leading to the fact that water OH bonds are fixed in classical FFs. This approach neglects vibrational information, limiting the possibility to study vibrational spectra directly. One solution is to estimate vibrational frequencies from MD simulation configurations without relying on explicit dynamics or vibrational modes. Skinner and co-workers developed a method using quadratic functions of the electric field on water OH bonds to predict vibrational properties.²⁷³ This approach was applied to predict infrared (IR), Raman, and sum frequency generation (SFG) spectra for bulk water,²⁷⁴ ionic aqueous solutions,²⁷⁵ and water/air interfaces.²⁷⁶ The model has since been extended to estimate the vibrations of amine groups in proteins, demonstrating its versatility for broader applications.²⁷⁷ Modern ML methods offer two new directions to enhance vibrational predictions from MD simulations. The first approach involves developing ML models to estimate vibrational properties directly from classical MD results. Rather than relying on simple electric field descriptors, molecules are represented by low-dimensional descriptors based on local structure, such as the ACFFs or geometric descriptors, similar to the strategy used in MLFFs. Various ML models, including principal component analysis, neural networks, and random forests, are applied as regression tools to map structural descriptors to vibrational frequencies, dipole moments, or polarizabilities, which can then be used to calculate vibrational spectra.²⁷⁸ For instance, Kananenka and co-workers used a Gaussian regression model to predict the vibrational properties of bulk water. Compared to the quadratic polynomial methods, their protocol reduced the RMSE by a factor of 2 while maintaining reasonable computational efficiency.²⁷⁹ To estimate the vibration of macromolecules such as proteins, DFT-level calculations can be performed on smaller molecular fragments. ML models are then employed to identify configurations of small molecules that correspond to specific protein environments. This approach allows properties of macromolecules to be inferred from their monomer unit, significantly reducing computational costs.²⁸⁰

Another approach for vibrational spectra prediction is calculating spectra directly from MD trajectories that include vibrational information. Dipole moments and polarizabilities can be estimated from simulation trajectories at DFT-level accuracy, using classical methods, such as assigning charges to

atomic centers, or by employing ML models to learn the polarizabilities.²⁸¹ Some MLFF frameworks are capable of simultaneously generating force fields and polarizabilities of molecules, enabling the calculation of IR and Raman spectra directly.^{128,282,283} At interfaces, SFG spectra are mainly contributed from the interfacial region but not the bulk water, containing surface specific structural and dynamics information.^{25,284} At water/air interfaces, where modeling the air phase is unnecessary, SFG spectra have been calculated and compared with results from DFT-MD simulations.^{285,286} Selloni and co-workers calculated the SFG spectra of water near anatase (101) surfaces, revealing how water coverage affects binding to TiO₂ surfaces, which is challenging to study experimentally.¹⁵⁹ The calculation of water vibrational spectra also extends to ice.²⁸⁷ Mauri and co-workers examined vibrational spectra for ice phases VIII and X, demonstrating that NQEs reduce the critical pressure for phase transitions between these phases.²⁸⁸ Tang and co-workers investigated water/alumina interfaces using MLFFs, calculating IR, Raman, and SFG spectra. They employed both dipole–polarizability correlation functions ($\mu - \alpha$ approach, Figure 14a), derived from MLFF modules, and

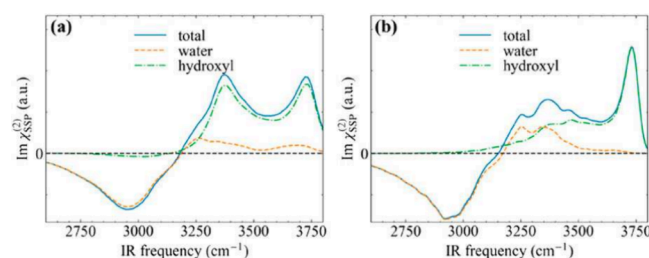


Figure 14. Imaginary part of SFG spectra of interfacial water and alumina surface OH groups calculated from MD simulations based on MLFFs using the (a) μ – α approach or (b) surface-specific VVCF approach. Figures are reprinted from ref.²⁸⁹ (Copyright 2024, AIP).

velocity–velocity correlation functions (VVCF, Figure 14b), obtained directly from simulation trajectories.²⁸⁹ The ML-based methods for calculating vibrational spectra have shown sufficient accuracy and are unaffected by proton transfer in water, making them highly versatile. These approaches expand the applicability of vibrational SFG calculations, enabling a deeper understanding of water interfaces and aiding the interpretation of nonlinear optical experimental observations.

5. OUTLOOK AND PERSPECTIVE

In this review, we discussed how MD simulations with ML enhanced sampling and MLFFs enable the investigation of complex physical and chemical processes in bulk and interfacial aqueous systems.

MLFFs enable MD simulations to achieve high-level quantum accuracy at a computational cost comparable to that of classical MD. This advancement has significantly expanded the scope of MD simulations, allowing researchers to tackle problems that were previously difficult to study, including but not limited to the calculation of free energy surfaces for chemical reactions and vibrational spectra. However, several challenges remain in this field:

1. Accuracy of Training Data: Most MLFFs are trained on data from density functional theory molecular dynamics simulations. However, the accuracy of the MLFFs is inherently limited by the accuracy of the DFT functional

used for training. If the underlying DFT calculations contain systematic errors, these inaccuracies will propagate into the MLFFs. For instance, dispersion interaction is necessary to improve the accuracy of DFT simulations of water, and MLFFs with such interactions included could improve the prediction of water density and melting point.^{290,291}

2. MLFFs are typically trained on thousands of configurations generated from DFT-MD simulations. Ideally, the training data set should include all possible configurations sampled during the production run with MLFFs. Currently, enhanced sampling methods have been incorporated into DFT-MD simulations. However, given the limited time scales of DFT-MD, training data for MLFFs must be carefully prepared to cover as broad a configurational space as possible.
3. While MLFFs are highly effective at capturing short-range interactions, they often struggle with long-range interactions, which are crucial for accurately modeling interfaces.
4. Early MLFFs were tested for relatively simple systems with a limited number of chemical elements. In the future, MLFFs must be adapted to handle increasingly complex multielement systems. Similar to classical FFs, the transferability and compatibility of MLFFs need more attention, particularly in simulations involving water/solid interfaces. Recently, MLFFs based on the LassoLars algorithm have been developed, demonstrating improved transferability and extrapolation performance for configurations outside the training set when compared with neural network potentials.²⁹²

Many current enhanced sampling methods rely on the selection of CVs, which are derived from structural descriptors obtained from MD simulations. Typically, a limited number of descriptors are selected for analysis using ML models, as including all possible structural features for large systems is computationally impractical. These selected descriptors are often based on human intuition, meaning the ML models can only analyze features explicitly included in the CVs. For instance, in the nucleation of NaCl from aqueous solutions, the removal of solvent water is a key driver of nucleation. However, if solvent effects are excluded when designing the CV, ML models may incorrectly assign the assembly of counterions as the primary feature driving nucleation.^{53,247} While this interpretation is reasonable within the given model, since the removal of solvent creates cavities filled by counterion assemblies, it fails to capture the crucial role of solvent water. To address this limitation, one solution is to develop CVs that can automatically capture complex structural features, such as those derived from GNN-based CVs.^{293,294}

Throughout this review, we highlighted numerous MD studies utilizing MLFFs or ML-enhanced sampling methods. One of the most powerful applications of these tools is estimating free energy surfaces for complex processes. However, most studies currently employ only one ML-based approach, either MLFFs or enhanced sampling. Combining these methods in future research holds great potential for providing even deeper insights into complex systems. We anticipate that such combined simulations will become more prevalent as ML techniques continue to advance. In addition, most enhanced sampling methods focused on calculating free energies, whereas calculating mechanistic and kinetic information remains

challenging because transition path sampling can be more expensive. As a result, continued method development for mechanistic studies of rare events in aqueous, interfacial systems is still needed.

6. SUMMARY

To summarize, here we have discussed the role of ML in MD simulations of aqueous, interfacial systems. We focused on the necessity of ML in this active and crucial research area, the advancements brought by ML-enhanced sampling methods, the development of ML force fields, and the diverse applications of ML in MD simulations.

We discussed five key topics in the application of ML-enhanced MD simulations. ML-enhanced MD simulations provide more accurate predictions of the structure and dynamics of bulk water and water interfaces, capturing subtle behaviors that are challenging for traditional methods. With ML-based MD simulations, proton transfer processes can now be studied over longer time scales and larger simulation systems, allowing researchers to investigate the effects of solvent structure in unprecedented detail. The combination of ML-enhanced sampling and MLFFs enables feasible calculations of reaction barriers for catalytic processes and phase transitions. By incorporating transition path sampling and free energy surface calculations, ML-enhanced MD simulations can also quantify reaction pathways with high accuracy. MLFFs facilitate large and long simulations that are essential for accurately calculating vibrational spectra, aligning computational results with experimental measurements and improving the interpretation of experimental data. While ML-enhanced MD simulations are powerful, there are areas for improvement as well. For instance, in order to apply dimensionality reduction methods to extract RCs for enhanced sampling MD simulations, a list of order parameters need to be predefined as possible components of RCs. However, these OPs most likely depend on human intuition, and dimensionality reduction methods generally cannot yet find completely orthogonal missing OPs. Additionally, integrating ML-enhanced sampling with MLFFs can expand the range of solvable problems, offering deeper insights into complex systems. Machine learning has revolutionized MD simulations, enabling the study of complex aqueous and interfacial systems with unprecedented accuracy and efficiency. As ML methodologies continue to evolve, we anticipate the development of new tools and combined approaches that will address existing challenges and open up new frontiers in the simulation of aqueous, interfacial systems.

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Notes

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